

MGT7179 Assignment 1

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INTRODUCTION

Today, companies are dealing with different kinds of customers and using fancy technology like data analysis and machine learning to understand them better, especially when it comes to important stuff like how well products work and how safe they are. These tools help figure out what customers like, how they behave, and what problems they might have, so businesses can make smarter decisions about how to make their products better. It's important to recognize that customers are all different, especially in industries like healthcare, where safety is super important, so companies can make products that fit their needs.

New improvements in analyzing data make it easier to group customers and predict how well products will do, making experiences safer and more personal. This report looks at how data analysis helps companies deal with the differences among customers in how they use products and stay safe, with the aim of making products better and keeping customers happy and healthy.

```
#Install packages and load necessary libraries:
options(repos = c(CRAN = "https://cran.rstudio.com/"))

install.packages("tidyverse")

## Installing package into 'C:/Users/40385187/AppData/Local/R/win-
library/4.3'
## (as 'lib' is unspecified)

## package 'tidyverse' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\40385187\AppData\Local\Temp\Rtmp44fkEr\downloaded_packages

library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.3.3
## Warning: package 'ggplot2' was built under R version 4.3.3
## Warning: package 'dplyr' was built under R version 4.3.3

## — Attaching core tidyverse packages ————— tidyverse
2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.4
## ✓ forcats   1.0.0      ✓ stringr    1.5.0
## ✓ ggplot2    3.5.0      ✓ tibble     3.2.1
```

```

## ✓ lubridate 1.9.2      ✓ tidyr      1.3.0
## ✓ purrr      1.0.1

## — Conflicts —————
tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()      masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to force
all conflicts to become errors

install.packages("caret")

## Installing package into 'C:/Users/40385187/AppData/Local/R/win-
library/4.3'
## (as 'lib' is unspecified)

## package 'caret' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\40385187\AppData\Local\Temp\Rtmp44fkEr\downloaded_packages

install.packages("MASS")

## Installing package into 'C:/Users/40385187/AppData/Local/R/win-
library/4.3'
## (as 'lib' is unspecified)

## package 'MASS' successfully unpacked and MD5 sums checked
## Warning: cannot remove prior installation of package 'MASS'

## Warning in file.copy(savedcopy, lib, recursive = TRUE): problem copying
## C:\Users\40385187\AppData\Local\R\win-
library\4.3\00LOCK\MASS\libs\x64\MASS.dll
## to C:\Users\40385187\AppData\Local\R\win-
library\4.3\MASS\libs\x64\MASS.dll:
## Permission denied

## Warning: restored 'MASS'

##
## The downloaded binary packages are in
## C:\Users\40385187\AppData\Local\Temp\Rtmp44fkEr\downloaded_packages

install.packages("dplyr")

## Warning: package 'dplyr' is in use and will not be installed

install.packages("car")

```

```
## Installing package into 'C:/Users/40385187/AppData/Local/R/win-  
library/4.3'  
## (as 'lib' is unspecified)  
  
## package 'car' successfully unpacked and MD5 sums checked  
##  
## The downloaded binary packages are in  
## C:\Users\40385187\AppData\Local\Temp\Rtmp44fkEr\downloaded_packages  
  
library(car)  
  
## Warning: package 'car' was built under R version 4.3.3  
  
## Loading required package: carData  
##  
## Attaching package: 'car'  
##  
## The following object is masked from 'package:dplyr':  
##  
##      recode  
##  
## The following object is masked from 'package:purrr':  
##  
##      some  
  
library(dplyr)  
library(MASS)  
  
## Warning: package 'MASS' was built under R version 4.3.3  
  
##  
## Attaching package: 'MASS'  
##  
## The following object is masked from 'package:dplyr':  
##  
##      select  
  
library(caret)  
  
## Warning: package 'caret' was built under R version 4.3.3  
  
## Loading required package: lattice  
##  
## Attaching package: 'caret'  
##  
## The following object is masked from 'package:purrr':  
##  
##      lift  
  
install.packages("ggplot2")  
  
## Warning: package 'ggplot2' is in use and will not be installed
```

```

library(ggplot2)
install.packages('tinytex')

## Installing package into 'C:/Users/40385187/AppData/Local/R/win-
library/4.3'
## (as 'lib' is unspecified)

## package 'tinytex' successfully unpacked and MD5 sums checked
##
## The downloaded binary packages are in
## C:\Users\40385187\AppData\Local\Temp\Rtmp44fkEr\downloaded_packages

tinytex::install_tinytex(force = TRUE) # install TinyTeX

## tlmgr install tlgpg
## tlmgr update --self
## tlmgr install tlgpg
## tlmgr --repository http://www.preining.info/tlgpg/ install tlgpg
## tlmgr option repository
"https://mirror.ox.ac.uk/sites/ctan.org/systems/texlive/tlnet"
## tlmgr update --list

```

Import the dataset

```

# Load the allocated dataset:
new_data <- read.csv("Q:/2024-02-17-Finall/2024-02-17-
Finall/OutputV01/output_40385187.csv")

```

Data Overview

For this study, I compiled a special dataset by incorporating student numbers and making specific code adjustments to make sure it fits my goals. The dataset contains 1,422,001 entries and 77 columns, providing detailed information about various aspects of product usage events.

```

dim(new_data)

## [1] 1422001      77

```

DATA PRE-PROCESSING AND EXPLORATION

Zero Values Analysis

- Examination of the “last year” Columns

This particular subset comprises rows where all columns beginning with “last_year” contain zero values. The resulting plot illustrates the frequency of various consequences, such as “Death,” “Injury,” “Malfunction,” and “No answer provided,” within the filtered dataset.

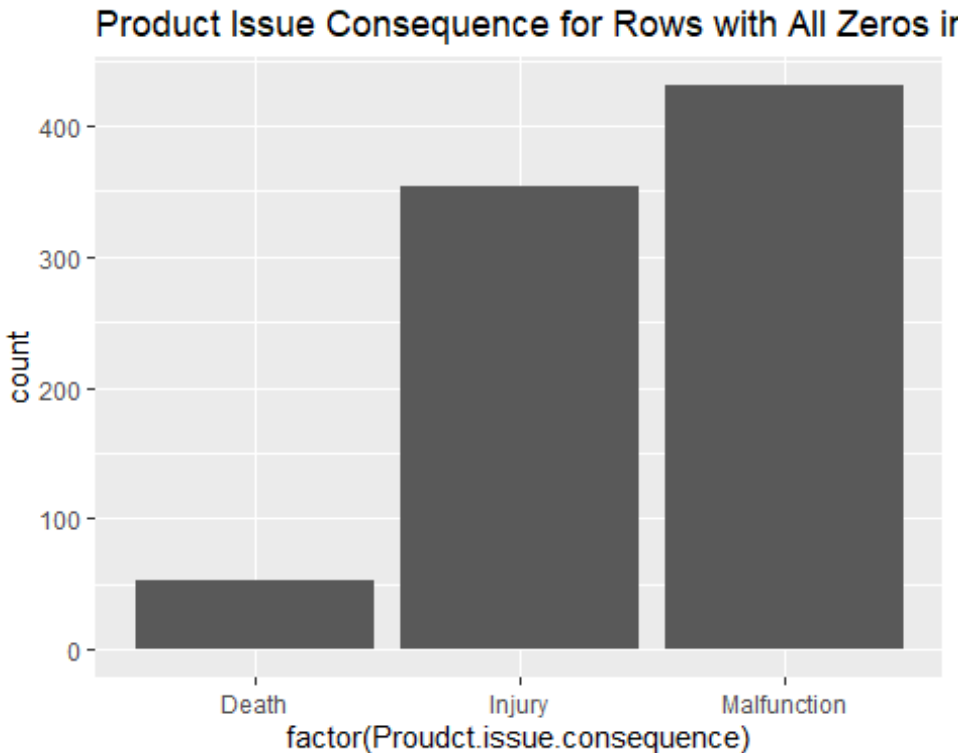
The plot reveals that “Malfunction” registers the highest count, implying it’s the most prevalent issue consequence reported in rows where all “last year...” columns contain zeros. This observation hints at a potential pattern or trend where certain types of product issues, notably malfunctions, might not be adequately captured by the annual metrics in the dataset or may be occurring in products that haven’t been reported frequently within the past year.

```
### analyzing "first year" variable subset
# Subset data for analysis: focusing on "last year..." category columns
last_year_columns <- new_data %>%
  dplyr::select(starts_with("last_year"))

# Check for rows where all "last year..." columns have zero values
rows_with_all_zeros_last_year <- which(rowSums(last_year_columns != 0) == 0)

# Extracting those rows for further analysis
new_data_rows_with_zeros_last_year <- new_data[rows_with_all_zeros_last_year,
]

# analyze the "Product.issue.consequence" for these specific rows:
if(!is.null(new_data_rows_with_zeros_last_year) &&
  nrow(new_data_rows_with_zeros_last_year) > 0) {
  ggplot(new_data_rows_with_zeros_last_year,
    aes(x=factor(Product.issue.consequence))) +
    geom_bar() +
    ggtitle("Product Issue Consequence for Rows with All Zeros in 'Last
Year...' Columns")
} else {
  print(" ")
}
```



- Examining “Last Two Year” Columns

The analysis narrows down the dataset to entries where all columns labeled “last two years...” contain zero values. Subsequently, it generates a bar plot depicting the distribution of “Proudct.issue.consequence.” Once again, “Malfunction” emerges as the most frequent issue, followed by “Injury,” with “Death” trailing behind. Notably, the absence of “Other” outcomes implies their non-existence within this subset of data. The prevalence of “Malfunction” hints that its causes may not be linked to factors tracked over the past two years, highlighting areas warranting further exploration in product monitoring and issue reporting.

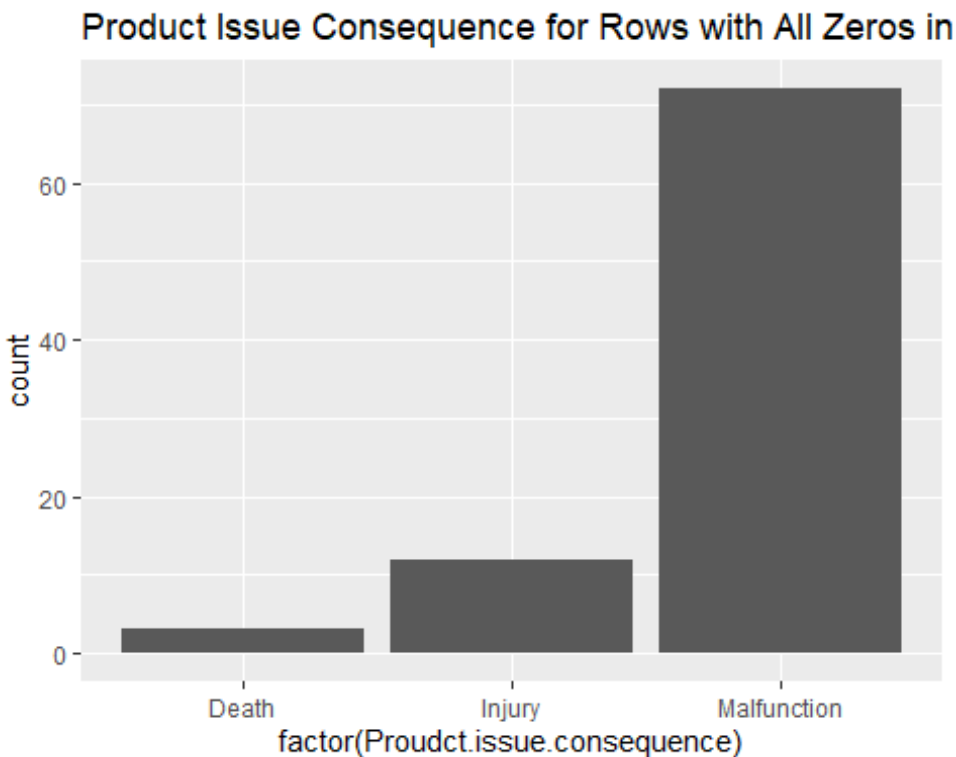
```
### analyzing "second year variable subset"
# Subset data for analysis: focusing on "last_two_years..." category columns
last_two_years_columns <- new_data %>%
  dplyr::select(starts_with("last_two_years"))

# Check for rows where all "last year..." columns have zero values
rows_with_all_zeros_last_two_years <- which(rowSums(last_two_years_columns !=
0) == 0)

# Extracting those rows for further analysis
new_data_rows_with_zeros_last_two_years <-
new_data[rows_with_all_zeros_last_two_years, ]

# analyze the "Proudct.issue.consequence" for these specific rows:
if(!is.null(new_data_rows_with_zeros_last_two_years) &&
```

```
nrow(new_data_rows_with_zeros_last_two_years) > 0) {
  ggplot(new_data_rows_with_zeros_last_two_years,
    aes(x=factor(Proudct.issue.consequence))) +
    geom_bar() +
    ggtitle("Product Issue Consequence for Rows with All Zeros in 'Last Two
Years...' Columns")
} else {
  print(" ")
}
```



- Examination of “Last Four Year” Columns

In our investigation spanning a four-year duration, we targeted columns labeled with “last_four_years” and aimed to identify instances with solely zero values, denoting a lack of recorded activity or events during that period. Our analysis revealed no such rows, as indicated by the absence of any message. This implies that within the context of the past four years, every row in the dataset exhibited some level of activity, suggesting consistent engagement or reporting throughout that timeframe.

```
### analyzing "fourth year variable subset"
# Subset data for analysis: focusing on "last_four_years..." category columns
last_four_years_columns <- new_data %>%
  dplyr::select(starts_with("last_four_years"))

# Check for rows where all "last four years..." columns have zero values
rows_with_all_zeros_last_four_years <- which(rowSums(last_four_years_columns
!= 0) == 0)
```

```

# Extracting those rows for further analysis
new_data_rows_with_zeros_last_four_years <-
new_data[rows_with_all_zeros_last_four_years, ]

# analyze the "Proudct.issue.consequence" for these specific rows:
if(!is.null(new_data_rows_with_zeros_last_four_years) &&
nrow(new_data_rows_with_zeros_last_four_years) > 0) {
  ggplot(new_data_rows_with_zeros_last_four_years,
aes(x=factor(Proudct.issue.consequence))) +
    geom_bar() +
    ggtitle("Product Issue Consequence for Rows with All Zeros in 'Last Four
Years...' Columns")
} else {
  print(" ")
}

## [1] " "

```

- Summary table

The summary table indicates that during the last year, 837 rows showed no recorded events, whereas only 87 rows remained inactive over the past two years. Interestingly, there were no instances of complete inactivity over the last four years, suggesting ongoing events or reporting within that timeframe.

```

# the data subsets:new_data_rows_with_zeros_last_year,
new_data_rows_with_zeros_last_two_years,
new_data_rows_with_zeros_last_four_years

# Count the number of rows for each category
num_rows_with_zeros_last_year <- nrow(new_data_rows_with_zeros_last_year)
num_rows_with_zeros_last_two_years <-
nrow(new_data_rows_with_zeros_last_two_years)
num_rows_with_zeros_last_four_years <-
nrow(new_data_rows_with_zeros_last_four_years)

# Create a summary table
summary_table <- data.frame(
  Category = c("Last Year", "Last Two Years", "Last Four Years"),
  Number_of_Rows_with_All_Zeros = c(num_rows_with_zeros_last_year,
num_rows_with_zeros_last_two_years, num_rows_with_zeros_last_four_years)
)

# View the summary table
print(summary_table)

##           Category Number_of_Rows_with_All_Zeros
## 1           Last Year                      837

```

## 2 Last Two Years	87
## 3 Last Four Years	0

- Categorical to factors:

In the dataset, all categorical variables have been transformed into factors to facilitate appropriate statistical analysis and modeling. Furthermore, this transformation is deemed necessary to ensure that statistical machine learning models can be run properly.

```
# Creating factors for all catagorical variables:
new_data$ID_non_uniq <- as.factor(new_data$ID_non_uniq)
new_data$Proudct.issue.consequence <-
as.factor(new_data$Proudct.issue.consequence)
new_data$manufacturer_contact_address_1 <-
as.factor(new_data$manufacturer_contact_address_1)
new_data$product.brand_name <- as.factor(new_data$product.brand_name)
new_data$product.generic_name <- as.factor(new_data$product.generic_name)
new_data$product.issue.type <- as.factor(new_data$product.issue.type)
new_data$type_of_report.1 <- as.factor(new_data$type_of_report.1)
new_data$reporter_job_code <- as.factor(new_data$reporter_job_code)
new_data$source_type <- as.factor(new_data$source_type)
new_data$product.manufacturer_name <-
as.factor(new_data$product.manufacturer_name)
new_data$product.product_operator <-
as.factor(new_data$product.product_operator)
new_data$product.manufacturer_city <-
as.factor(new_data$product.manufacturer_city)
new_data$product.manufacturer_state <-
as.factor(new_data$product.manufacturer_state)
new_data$product.manufacturer_country <-
as.factor(new_data$product.manufacturer_country)
new_data$product.field_description <-
as.factor(new_data$product.field_description)
new_data$product.product_report_product_code <-
as.factor(new_data$product.product_report_product_code)
```

- Graphical Analysis: Plot 1 The distribution of product issue consequences, specifically “Injury”, for the ‘General Hospital’ product field, across the top 10 manufacturer states, is presented in the plot. The data appears to have been filtered to display only one consequence, “Injury”, represented by a singular bar in the plot. The count for “Injury” remains consistent across all displayed manufacturer states, with no discernible variation. This uniform distribution could suggest either a lack of diversity in the provided data for this category or a potential issue with data visualization where states are not clearly differentiated on the x-axis. Further investigation is necessary to ascertain the reason behind this consistent distribution.

Plot 2 The focus of the code and plot is on product issues within the ‘Unknown’ category, likely encompassing unclassified products or those lacking specific category data. The plot reveals:

'Malfunction' as the most frequent issue, significantly outnumbering 'Injury'. Occurrences of 'Death' are minimal compared to other issue consequences. There are negligible counts for 'No answer provided' and 'Other', indicating that most issues are classified as either 'Malfunction' or 'Injury'. This pattern suggests that 'Malfunction' is a prevalent problem in the 'Unknown' category, underscoring the need for further investigation to uncover why malfunctions are notably common in these products.

Plot 3 The code narrows down to "Clinical Chemistry" products, identifying the top 10 states by occurrence. A bar plot displays "Proudct.issue.consequence" across these states, categorized into "Death" (red), "Injury" (green), and "Malfunction" (blue).

Key observations from the plot include:

"Malfunction" is the most frequently reported issue, with notable peaks in two states. "Injury" ranks second in frequency, evident but less prevalent than malfunctions. "Death" reports are the least frequent, suggesting their rarity compared to other consequences. This highlights "Malfunction" as the primary concern within "Clinical Chemistry", potentially due to product specifics or reporting habits. The lower incidence of "Death" could indicate effective safety measures or the non-lethal nature of reported issues. Geographical disparities point to potential trends in product issues or reporting practices, warranting further exploration.

Plot 4 The provided code filters the dataset for 'Cardiovascular' products and identifies the top 10 states by issue frequency. The plot categorizes 'Proudct.issue.consequence' into 'Death', 'Injury', and 'Malfunction', showcasing their distribution across these states.

Key observations include:

'Malfunction' is more prevalent than 'Death' and 'Injury' across all states. One state stands out with a notably higher count of 'Malfunction', closely followed by 'Injury'. 'Death' occurrences are the least frequent, with much lower frequencies compared to 'Injury' and 'Malfunction'. These trends highlight 'Injury' and 'Malfunction' as the primary concerns in cardiovascular product issues, particularly emphasizing the higher incidence of 'Injury'. This underscores the importance of investigating the causes and implementing preventive measures for these issues.

Plot 5 The code isolates "Anesthesiology" products, identifying the top 10 states by report volume. A bar plot displays "Proudct.issue.consequence," focusing on "Injury" and "Malfunction".

Plot findings:

The plot does not reveal the prevalent issue consequences for anesthesiology products, indicating critical areas for quality improvement and further research.

Plot 6 The code and accompanying plot present the distribution of product issue consequences—'Death', 'Injury', 'Malfunction', 'No answer provided', and 'Other'—across various product fields such as 'Anesthesiology', 'Cardiovascular', 'Clinical Chemistry', 'General Hospital', and 'Unknown'.

The faceted bar plot, segregating data by product field, reveals:

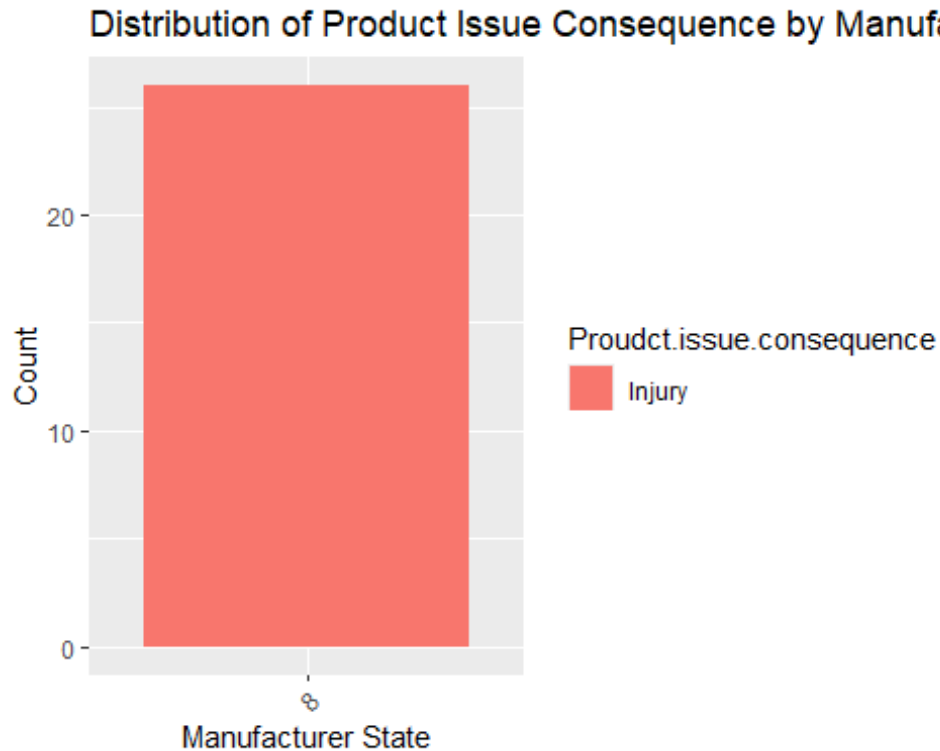
'Malfunction' is a prevalent issue, often dominating in proportion. 'Injury' closely follows, particularly notable in 'Cardiovascular' and 'Clinical Chemistry' fields. 'Death' constitutes a smaller fraction of reported issues compared to 'Malfunction' and 'Injury'. 'No answer provided' and 'Other' categories are less frequent across all fields. These findings illustrate the common occurrence and distribution of issue consequences within each product field, highlighting primary areas for potential safety enhancements and reliability improvements.

Graphical Analysis: whole original uploaded dataset: "new_data"

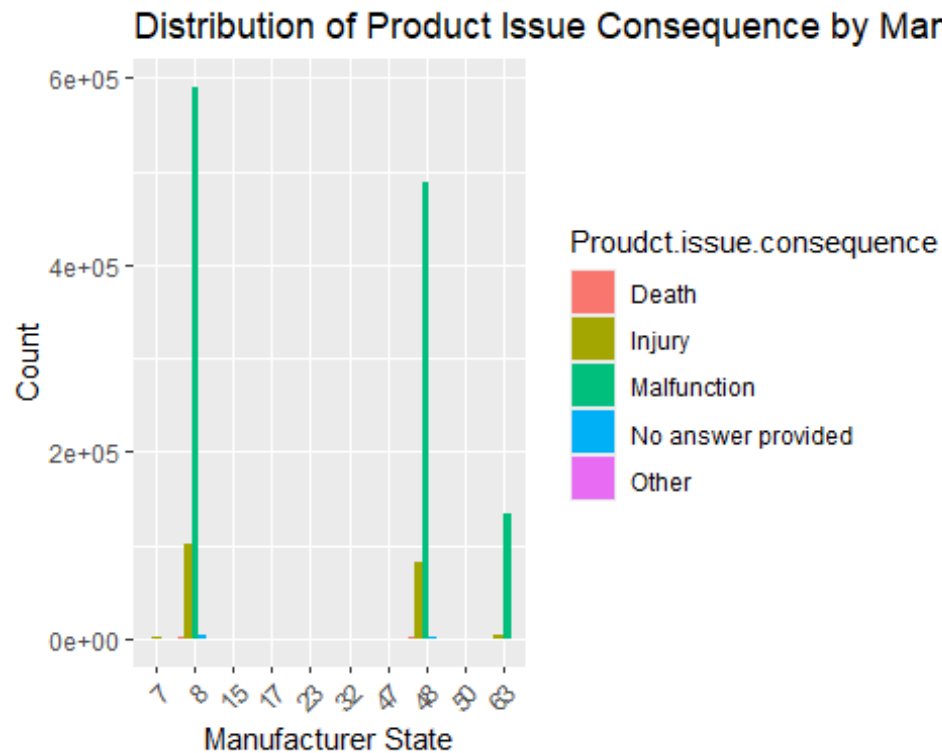
```
library(dplyr)
library(ggplot2)
```

Plots for Distribution of Product Issue Consequence by Manufacturer State for all 5 Product Fields present in my dataset.

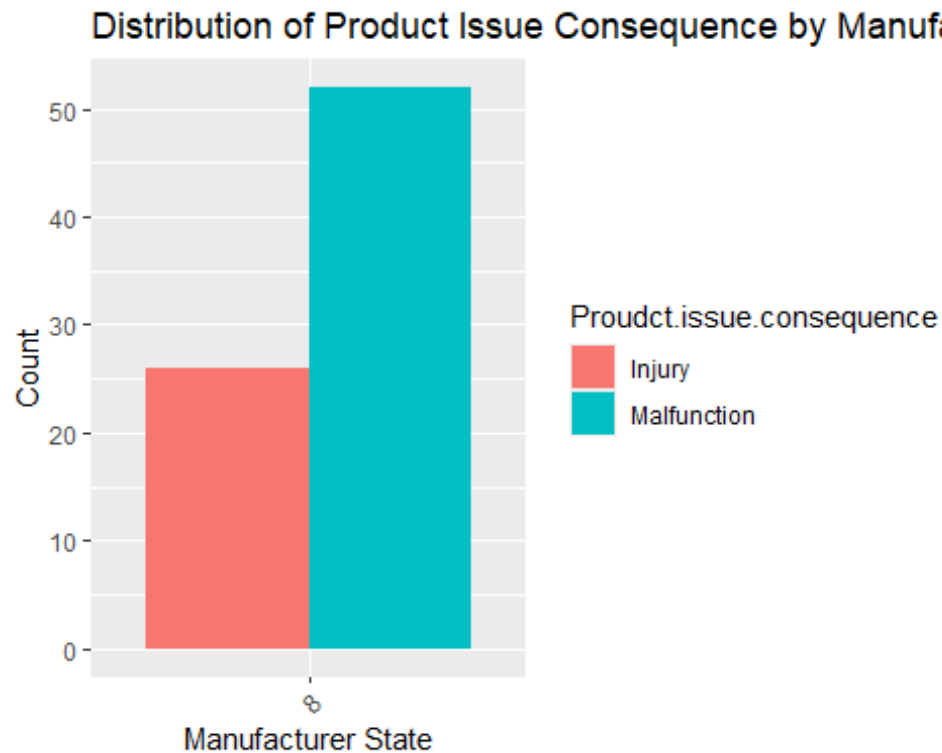
```
# Filter data for 'General Hospital' product field description
filtered_new_data_general_hospital <- new_data %>%
  filter(product.field_description == "General Hospital")
# focus on "product.manufacturer_state" as the predictor and find the top 10 states.
top_states_general_hospital <- filtered_new_data_general_hospital %>%
  count(product.manufacturer_state, sort = TRUE) %>%
  head(10)
# Plot 1: Distribution of "Proudct.issue.consequence" within "General Hospital" Category.
filtered_new_data_general_hospital %>%
  filter(product.manufacturer_state %in%
top_states_general_hospital$product.manufacturer_state) %>%
  ggplot(aes(x = product.manufacturer_state, fill =
Proudct.issue.consequence)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Product Issue Consequence by Manufacturer
State for General Hospital",
       x = "Manufacturer State",
       y = "Count") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



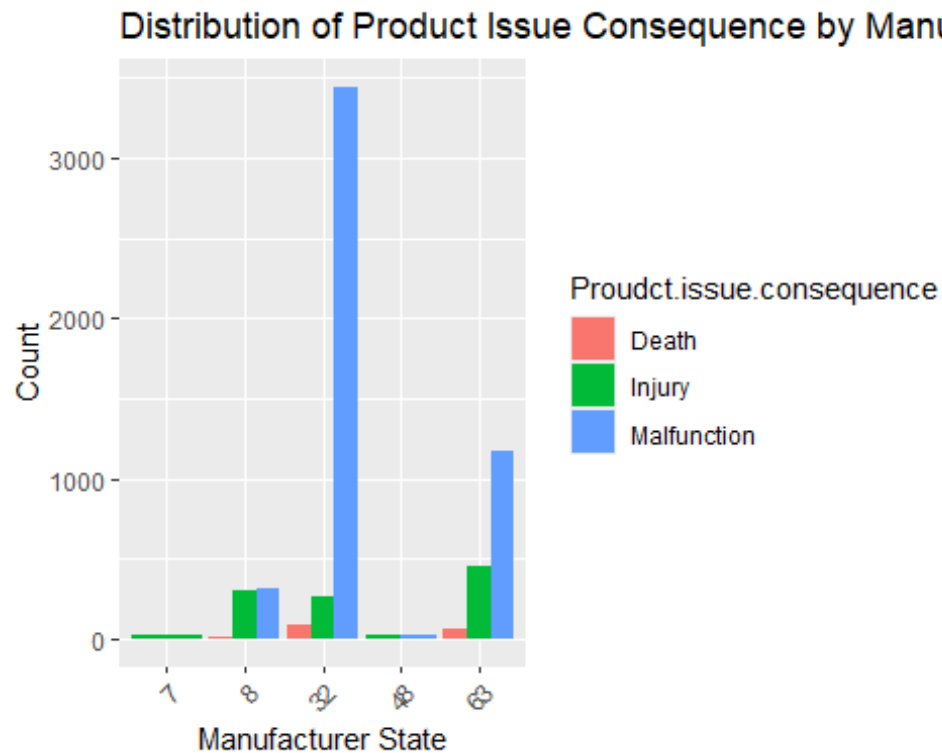
```
# Filter data for 'Unknown' product field description
filtered_new_data_unknown <- new_data %>%
  filter(product.field_description == "Unknown")
# identify the top 10 states in the "product.manufacturer_state" for the
"Unknown" product field.
top_states_unknown <- filtered_new_data_unknown %>%
  count(product.manufacturer_state, sort = TRUE) %>%
  head(10)
# Plot 2: Distribution of "Proudct.issue.consequence" for "Unknown" Category
filtered_new_data_unknown %>%
  filter(product.manufacturer_state %in%
top_states_unknown$product.manufacturer_state) %>%
  ggplot(aes(x = product.manufacturer_state, fill =
Proudct.issue.consequence)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Product Issue Consequence by Manufacturer
State for Unknown Product Field",
       x = "Manufacturer State",
       y = "Count") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# Filter data for 'Clinical Chemistry' product field description
filtered_new_data_clinical_chemistry <- new_data %>%
  filter(product.field_description == "Clinical Chemistry")
# identify the top 10 states within "product.manufacturer_state" for the
"Clinical Chemistry" product field.
top_states_clinical_chemistry <- filtered_new_data_clinical_chemistry %>%
  count(product.manufacturer_state, sort = TRUE) %>%
  head(10)
# Plot 3: Distribution of "Proudct.issue.consequence" for "Clinical
Chemistry" Category
filtered_new_data_clinical_chemistry %>%
  filter(product.manufacturer_state %in%
top_states_clinical_chemistry$product.manufacturer_state) %>%
  ggplot(aes(x = product.manufacturer_state, fill =
Proudct.issue.consequence)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Product Issue Consequence by Manufacturer
State for Clinical Chemistry",
       x = "Manufacturer State",
       y = "Count") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# Filter data for 'Cardiovascular' product field description
filtered_new_data_cardiovascular <- new_data %>%
  filter(product.field_description == "Cardiovascular")
# find the top 10 states in "product.manufacturer_state" for products
classified under "Cardiovascular".
top_states_cardiovascular <- filtered_new_data_cardiovascular %>%
  count(product.manufacturer_state, sort = TRUE) %>%
  head(10)
# Plot 4: Distribution of "Proudct.issue.consequence" for "Cardiovascular"
Category
filtered_new_data_cardiovascular %>%
  filter(product.manufacturer_state %in%
top_states_cardiovascular$product.manufacturer_state) %>%
  ggplot(aes(x = product.manufacturer_state, fill =
Proudct.issue.consequence)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Product Issue Consequence by Manufacturer
State for Cardiovascular",
       x = "Manufacturer State",
       y = "Count") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
# Filter data for 'Anesthesiology' product field description
filtered_new_data_anesthesiology <- new_data %>%
  filter(product.field_description == "Anesthesiology")
# identify the top 10 states within "product.manufacturer_state" for the
"Anesthesiology" product field.
top_states_anesthesiology <- filtered_new_data_anesthesiology %>%
  count(product.manufacturer_state, sort = TRUE) %>%
  head(10)
# Plot 5: Distribution of "Proudct.issue.consequence" for "Anesthesiology"
Category
filtered_new_data_anesthesiology %>%
  filter(product.manufacturer_state %in%
top_states_anesthesiology$product.manufacturer_state) %>%
  ggplot(aes(x = product.manufacturer_state, fill =
Proudct.issue.consequence)) +
  geom_bar(position = "dodge") +
  labs(title = "Distribution of Product Issue Consequence by Manufacturer
State for Anesthesiology",
    x = "Manufacturer State",
    y = "Count") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

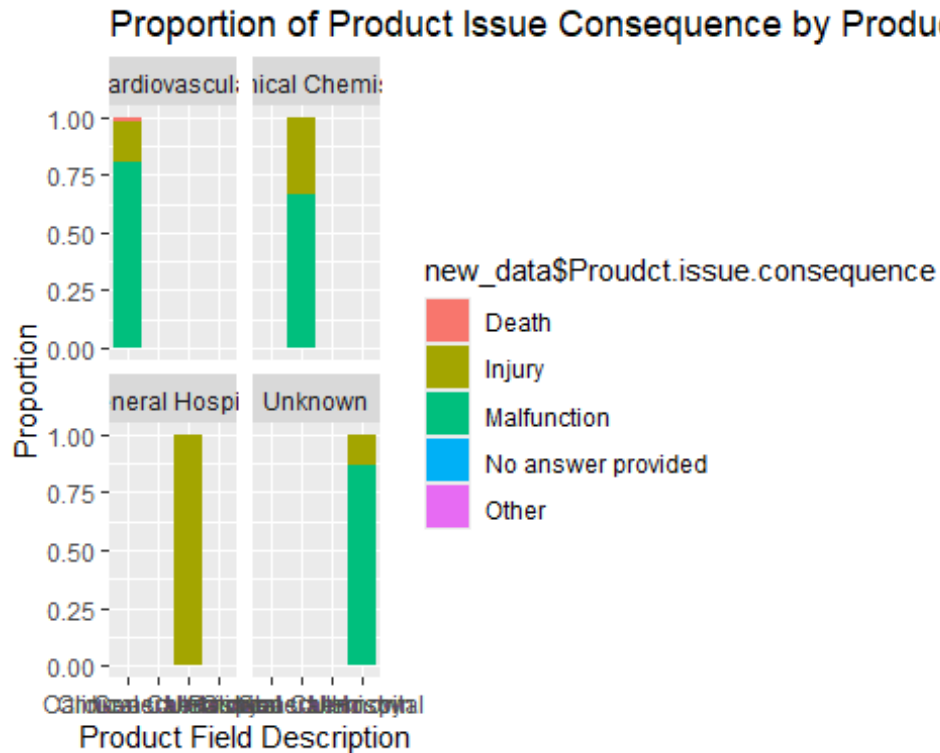
Distribution of Product Issue Consequence by Manufacturer



```
# Plot 6:
ggplot(new_data, aes(x = new_data$product.field_description, fill =
new_data$Proudct.issue.consequence)) +
  geom_bar(position = "fill") +
  facet_wrap(~ new_data$product.field_description) +
  labs(title = "Proportion of Product Issue Consequence by Product Field
Description",
        x = "Product Field Description", y = "Proportion")

## Warning: Use of `new_data$product.field_description` is discouraged.
## [i] Use `product.field_description` instead.

## Warning: Use of `new_data$Proudct.issue.consequence` is discouraged.
## [i] Use `Proudct.issue.consequence` instead.
```



- Data Processing

The code outlines data preparation steps for analysis, with a focus on streamlining and simplification:

Dropping Irrelevant Columns: The code removes `ID_non_uniq` and `date_event` from the `new_data` dataframe to create `dataframe`. This step suggests that these columns are deemed irrelevant for predictive purposes or lack relevance, aiming to reduce complexity and concentrate on meaningful data.

Binary Encoding of Target Variable: The target variable `Proudct.issue.consequence` is binary encoded, with “Malfunction” assigned as 1 and other outcomes as 0. This simplifies the analysis into a binary classification task, streamlining the focus on identifying malfunctions.

Creating a Manageable Subset: To address computational limits, a subset comprising the first 60,000 rows is created. This balances the utilization of data with the capabilities of the system, ensuring that the dataset remains manageable for analysis without causing system overload.

Additionally, a `table` command likely checks the distribution of the binary encoding, confirming the successful preparation of data for focused and efficient analysis within computational constraints.

```
# Creating dataset for analysis by dropping ID-non_uniq and date_event
because they are irrelevant
dataframe <- new_data %>%
```

```

dplyr::select(-ID_non_uniq, -date_event)

# Malfunction is dominant class, so malfunction =1 , all others = 0
dataframe <- dataframe %>%
  mutate(Proudct.issue.consequence = ifelse(Proudct.issue.consequence ==
"Malfunction", 1, 0))
table(dataframe$Proudct.issue.consequence)

##
##      0      1
## 202894 1219107

# New dataset with first 60,000 observations(rows)
altered_new_data <- dataframe[1:60000, ]

dummy_var_new_data <- altered_new_data

```

- Creating Dummy Variables for Lasso Feature Selection

Utilizing dummy variables is vital for advanced feature selection methods like LASSO regression, especially when handling categorical data. This procedure converts categorical variables into binary ones, which are then utilized in regression analysis. In this scenario, 'Proudct.issue.consequence' continues to serve as a factor to differentiate 'Malfunction' from other outcomes, aligning well with the binary nature of logistic regression.

Through the exclusion of 'ID_non_uniq', irrelevant data is eliminated, allowing the focus to remain on influential variables. This preprocessing step is critical as it empowers LASSO regression to effectively tackle challenges such as data dimensionality and multicollinearity. Additionally, it improves model interpretability by evaluating the impacts of various category levels on the target variable.

In essence, this approach fortifies the predictive modeling process, fostering deeper insights and yielding better model performance, thus enhancing comprehension of the data.

```

# Create dummy variables:
# Note: The response variable 'Proudct.issue.consequence' remains as a factor
# and is not dummy coded as it is the target variable for predictive
modelling.
# Furthermore, ID_non_uniq was excluded
install.packages("fastDummies")

## Installing package into 'C:/Users/40385187/AppData/Local/R/win-
library/4.3'
## (as 'lib' is unspecified)

## package 'fastDummies' successfully unpacked and MD5 sums checked
##

```

```
## The downloaded binary packages are in
## C:\Users\40385187\AppData\Local\Temp\Rtmp44fkEr\downloaded_packages

library(fastDummies)

## Warning: package 'fastDummies' was built under R version 4.3.3

## Thank you for using fastDummies!

## To acknowledge our work, please cite the package:

## Kaplan, J. & Schlegel, B. (2023). fastDummies: Fast Creation of Dummy
(Binary) Columns and Rows from Categorical Variables. Version 1.7.1. URL:
https://github.com/jacobkap/fastDummies,
https://jacobkap.github.io/fastDummies/.

dummy_var_new_data <- dummy_cols(dummy_var_new_data, select_columns =
c('manufacturer_contact_address_1', 'product.brand_name',
                                     'product.generic_name',
'product.issue.type', 'type_of_report.1',
                                     'reporter_job_code',
'source_type', 'product.manufacturer_name',
'product.product_operator', 'product.manufacturer_city',
'product.manufacturer_state',
'product.manufacturer_country', 'product.field_description',
'product.product_report_product_code'))

df <- dummy_var_new_data
```

- Creating Subsets

For feature selection using LASSO regression, we divided the data into four subsets based on 'product.manufacturer_state' and 'product.field_description'. This segmentation enhances the precision of analysis by accounting for variations specific to states and product fields.

Subset 1: State 8 is focused on due to its significant data volume, diverse range of issues, or business relevance, with the aim of analyzing product issues in this particular location.

Subset 2: State 48 is targeted for reasons similar to subset 1, recognizing unique state characteristics that influence patterns of product issues.

Subset 3: Records from states 7 , 32 , 47 , 50 and 63 are aggregated, providing a comprehensive comparison to capture diverse or representative trends across states.

Subset 4: Emphasis is placed on 'Unknown' product fields, delving into challenges related to data classification, reporting practices, or issues within undefined categories.

Following segmentation, the columns 'product.manufacturer_state' and 'product.field_description' are removed to optimize the application of LASSO. Since these variables remain constant within subsets, they do not contribute to enhancing model insight.

```
# Subset creation:

# Create subset1 containing rows where product.manufacturer_state == 8
subset1 <- df[df$product.manufacturer_state == 8, ]
subset1 <- subset1[, !names(subset1) %in% 'product.manufacturer_state']

# Create subset2 containing rows where product.manufacturer_state == 48
subset2 <- df[df$product.manufacturer_state == 48, ]
subset2 <- subset2[, !names(subset2) %in% 'product.manufacturer_state']

# Create subset3 containing rows where product.manufacturer_state matches multiple values
subset3 <- df[df$product.manufacturer_state %in% c(7, 32, 47, 50, 63), ]
subset3 <- subset3[, !names(subset3) %in% 'product.manufacturer_state']

# Create subset 4 which includes only product feild = "Unknown"
subset4 <- df[df$product.field_description == "Unknown",]
subset4 <- subset4[, !names(subset4) %in% 'product.field_description']
```

STATISTICAL ANALYSIS

Lasso Regression

- LASSO Subset 1: In our examination using LASSO regression on Subset 1, we encoded categorical variables and partitioned the dataset into training and test sets to ensure resilience. The objective was to pinpoint significant predictors for 'Proudct.issue.consequence' by delving into variations within the state, employing cross-validation to determine the optimal lambda value for minimizing prediction errors.

Key discoveries include:

Emphasis on critical predictors within state 8, identifying those with the most substantial impact on product issues, facilitated by LASSO's feature selection capability. The optimal lambda, identified through cross-validation, was $(3.528047 \times 10^{-5})$, underscoring the necessity of striking a balance between model complexity and predictive accuracy. Coefficient path plots illustrated how the influence of predictors changes with varying lambda values, highlighting variables that retain significance. The model achieved an accuracy rate of 79.6% on the test set, indicating robust predictive performance. Noteworthy predictors include 'manufacturer_contact_address_13016' with a negative coefficient, indicating its inverse relationship with malfunction likelihood, along with other variables suggesting moderate associations. This analysis underscores the effectiveness of LASSO regression in identifying meaningful predictors within intricate datasets, guiding

subsequent logistic regression and LDA analyses to deepen comprehension of product issue dynamics within specific contexts.

LASSO Subset 1

creating dummy variables of predictor variables

```
x <- model.matrix(Proudct.issue.consequence ~ ., subset1)[, -1]
```

separating response variable

```
y <- subset1$Proudct.issue.consequence
```

splitting train and test data

```
set.seed(40385187)
```

```
train.l = sample(1:nrow(x), nrow(x)/2)
```

```
test.l = (-train.l)
```

```
y.test.l = y[test.l]
```

creating a Lambda sequence

```
grid = 10^seq(10, -2, length = 100)
```

```
install.packages("glmnet") # Install the glmnet package
```

```
## Installing package into 'C:/Users/40385187/AppData/Local/R/win-  
library/4.3'
```

```
## (as 'lib' is unspecified)
```

```
## package 'glmnet' successfully unpacked and MD5 sums checked
```

```
##
```

```
## The downloaded binary packages are in
```

```
## C:\Users\40385187\AppData\Local\Temp\Rtmp44fkEr\downloaded_packages
```

```
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.3.3
```

```
## Loading required package: Matrix
```

```
##
```

```
## Attaching package: 'Matrix'
```

```
## The following objects are masked from 'package:tidyr':
```

```
##
```

```
## expand, pack, unpack
```

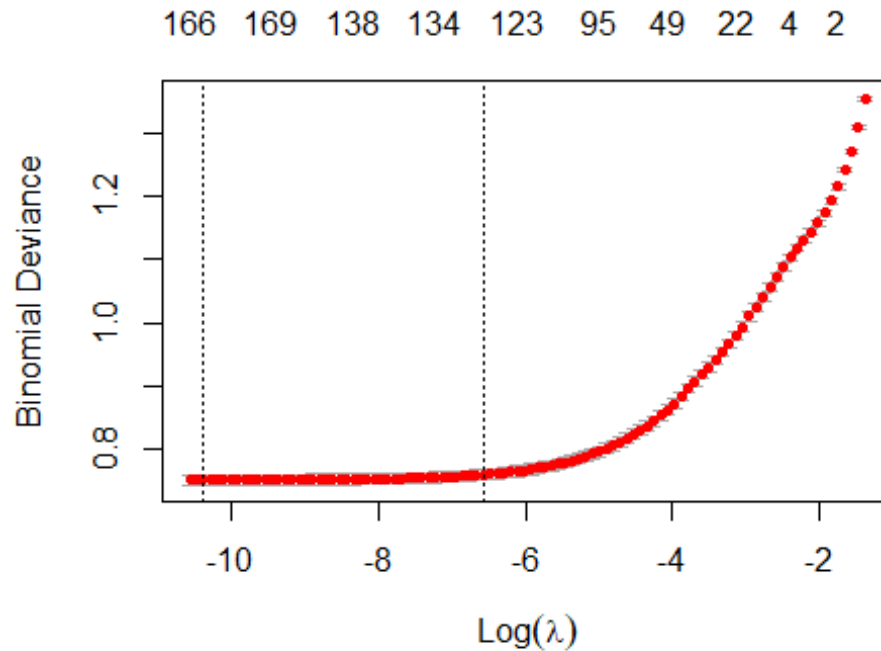
```
## Loaded glmnet 4.1-8
```

Fit Lasso model with cross-validation to find best Lambda

```
set.seed(40385187)
```

```
cv_modelA01 <- cv.glmnet(x[train.l,], y[train.l], family="binomial", alpha=1,  
type.measure = "deviance", nfolds =10)
```

```
# Plot the cross-validation error as a function of lambda
plot(cv_modelA01)
```

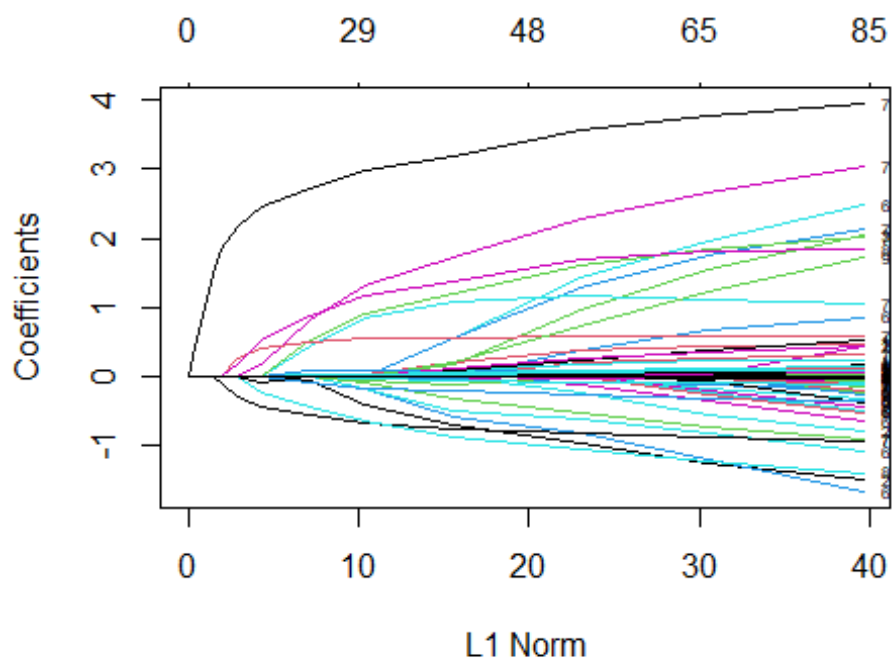


```
# Select the best lambda value
best_lambdaA01 <- cv_modelA01$lambda.min
best_lambdaA01

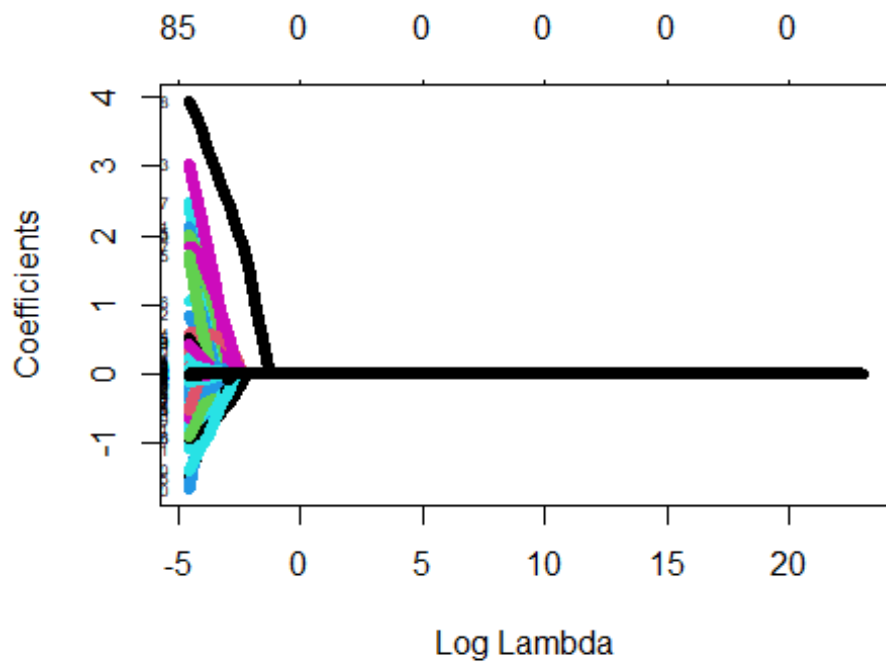
## [1] 3.077939e-05

# fitting lasso model on train set
set.seed(40385187)
lasso.mod=glmnet(x[train.1,], y[train.1],family="binomial",type.measure =
"deviance", alpha=1, lambda=grid)
plot(lasso.mod, label=TRUE)

## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```



```
plot(lasso.mod, xvar="lambda", label=TRUE, lwd=6)
```



```

# prediction using lasso
set.seed(40385187)
lasso.pred <- predict(lasso.mod,family="binomial", s = best_lambdaA01, newx =
x[test.1,])
pred_class <- ifelse(lasso.pred >= 0.5, 1, 0)
accuracy <- mean(pred_class == y[test.1])
accuracy

## [1] 0.796087

# running lasso on whole data set
set.seed(40385187)
out = glmnet(x, y, alpha=1, family="binomial", type.measure = "deviance",
lambda = grid)

# extracting coefficients of lasso model
set.seed(40385187)
lasso.coef=predict(out, type="coefficients", s=best_lambdaA01)[1:79,]
lc <- lasso.coef[lasso.coef!=0]

# tabulate coefficients
lc_df <- data.frame(variable = names(lc)[-1], coefficient = lc[-1])

# extracting top ten coefficients
top10 <- head(lc_df[order(abs(lc_df$coefficient), decreasing = TRUE), ], 10)
top10

##
variable
## manufacturer_contact_address_13016
manufacturer_contact_address_13016
## last_two_years_root_cause_description_most_freq
last_two_years_root_cause_description_most_freq
## last_two_years_decision_date_max_changes_in_product
last_two_years_decision_date_max_changes_in_product
## last_year_company_name_most_freq
last_year_company_name_most_freq
## last_two_years_all_product_codes_most_freq
last_two_years_all_product_codes_most_freq
## last_two_years_decision_date_average_changes_in_product
last_two_years_decision_date_average_changes_in_product
##
coefficient
## manufacturer_contact_address_13016 -1.649573e-01
## last_two_years_root_cause_description_most_freq -3.786729e-02
## last_two_years_decision_date_max_changes_in_product -1.401603e-02
## last_year_company_name_most_freq 7.447491e-04
## last_two_years_all_product_codes_most_freq -1.022622e-04
## last_two_years_decision_date_average_changes_in_product -1.111464e-15

```

- **LASSO Subset 2:** For Subset 2, we employed a LASSO regression approach similar to Subset 1, with a focus on identifying key predictors. Categorical variables were encoded, and the data was split for validation. Cross-validation was then utilized to determine the optimal lambda, yielding a value of approximately $(3.554902 \times 10^{-5})$. Plots were generated to visualize predictor selection and coefficient shrinkage with increasing regularization, aiding in lambda selection and highlighting significant predictors across different regularization levels.

The chosen lambda resulted in approximately 80.57% accuracy on the test set, indicating a robust model, albeit slightly less accurate compared to Subset 1. This performance suggests that the model captures a substantial portion of the variance in 'Proudct.issue.consequence', although other factors or data nuances may further elucidate product issue outcomes.

Key predictors for Subset 2 included recurring variables like 'last_two_years_all_product_codes_most_freq', as well as new indicators such as 'last_four_years_all_product_codes_most_freq' and 'last_year_all_product_codes_most_freq', underscoring the importance of manufacturing details and recent product alterations in predicting issues.

This analysis underscores the effectiveness of LASSO regression in feature selection and the insights it provides into critical factors influencing product issues within specific manufacturing contexts. These insights can guide targeted efforts for quality improvement.

LASSO Subset 2

```
# creating dummy variables of predictor variables
x2 <- model.matrix(Proudct.issue.consequence ~ ., subset2)[, -1]

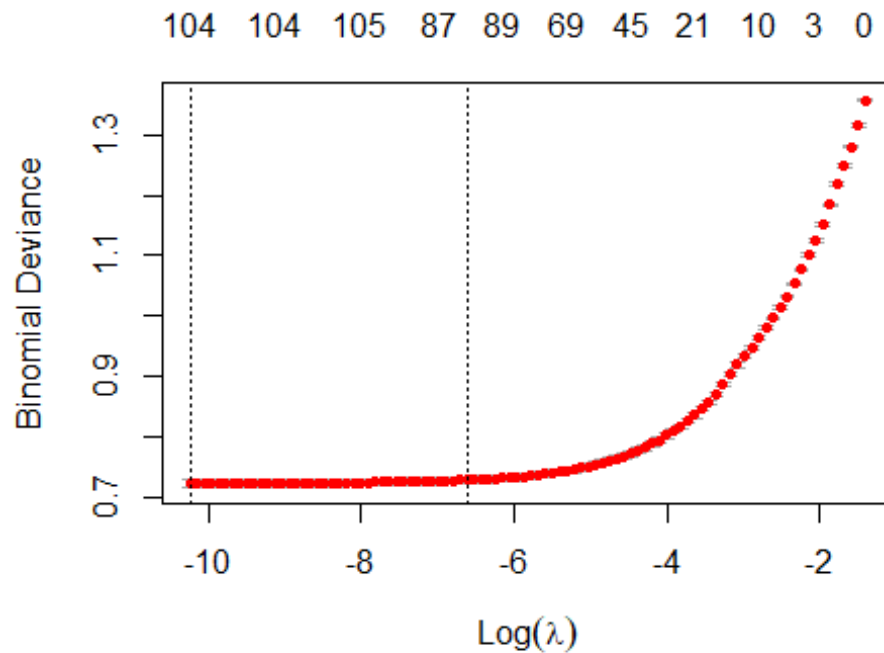
# separating response variable
y2 <- subset2$Proudct.issue.consequence

# splitting train and test data
set.seed(40385187)
train.l = sample(1:nrow(x2), nrow(x2)/2)
test.l = (-train.l)
y2.test.l = y2[test.l]

# creating a lambda sequence
grid2 = 10^seq(10, -2, length = 100)

#install.packages("glmnet") # Install the glmnet package
#library(glmnet)
# Fit Lasso model with cross-validation to find best Lambda
set.seed(40385187)
cv_modelA02 <- cv.glmnet(x2[train.l,], y2[train.l], family="binomial",
alpha=1, type.measure = "deviance", nfolds =10)
```

```
# Plot the cross-validation error as a function of Lambda
plot(cv_modelA02)
```



```
# Select the best lambda value
best_lambdaA02 <- cv_modelA02$lambda.min
best_lambdaA02

## [1] 3.554902e-05

# fitting lasso model on train set
set.seed(40385187)
lasso.mod=glmnet(x2[train.1,], y2[train.1],family="binomial",type.measure =
"deviance", alpha=1, lambda=grid2)
plot(lasso.mod, label=TRUE)

## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```



```

# prediction using lasso
set.seed(40385187)
lasso.pred2 <- predict(lasso.mod,family="binomial", s = best_lambdaA02, newx
= x2[test.1,])
pred_class2 <- ifelse(lasso.pred2 >= 0.5, 1, 0)
accuracy2 <- mean(pred_class2 == y2[test.1])
accuracy2

## [1] 0.8057318

# running lasso on whole data set
set.seed(40385187)
out = glmnet(x2, y2, alpha=1, family="binomial", type.measure = "deviance",
lambda = grid2)

# extracting coefficients of lasso model
set.seed(40385187)
lasso.coef=predict(out, type="coefficients", s=best_lambdaA02)[1:79,]
lc2 <- lasso.coef[lasso.coef!=0]

# tabulate coefficients
lc_df2 <- data.frame(variable = names(lc2)[-1], coefficient = lc2[-1])

# extracting top ten coefficients
top10_2 <- head(lc_df2[order(abs(lc_df2$coefficient), decreasing = TRUE), ],
10)
top10_2

##
variable
## last_year_all_product_codes_most_freq
last_year_all_product_codes_most_freq
## last_two_years_all_product_codes_most_freq
last_two_years_all_product_codes_most_freq
## last_four_years_all_product_codes_most_freq
last_four_years_all_product_codes_most_freq
##
## coefficient
## last_year_all_product_codes_most_freq -2.787017e-04
## last_two_years_all_product_codes_most_freq -2.547762e-18
## last_four_years_all_product_codes_most_freq -3.807000e-19

```

- LASSO Subset 3: In the LASSO regression analysis for Subset 3, we followed the established approach, employing cross-validation to determine the optimal lambda (0.003179135 0.003179135) for reducing complexity and conducting variable selection. This iterative process identifies the most influential predictors by shrinking less relevant coefficients towards zero.

The model achieved an impressive 91.67% accuracy on the test set, highlighting its robust performance. The analysis emphasizes the variables with the strongest impact on

'Proudct.issue.consequence', particularly emphasizing the significance of recent product changes within the last two years.

These critical predictors, identified through their coefficients, serve as valuable insights for further analysis and action. They enable stakeholders to pinpoint and address the key factors behind product issue outcomes, facilitating targeted decision-making by shedding light on the essential elements influencing product issues in this specific context.

LASSO Subset 3

```
# creating dummy variables of predictor variables
x3 <- model.matrix(Proudct.issue.consequence ~ ., subset3)[, -1]

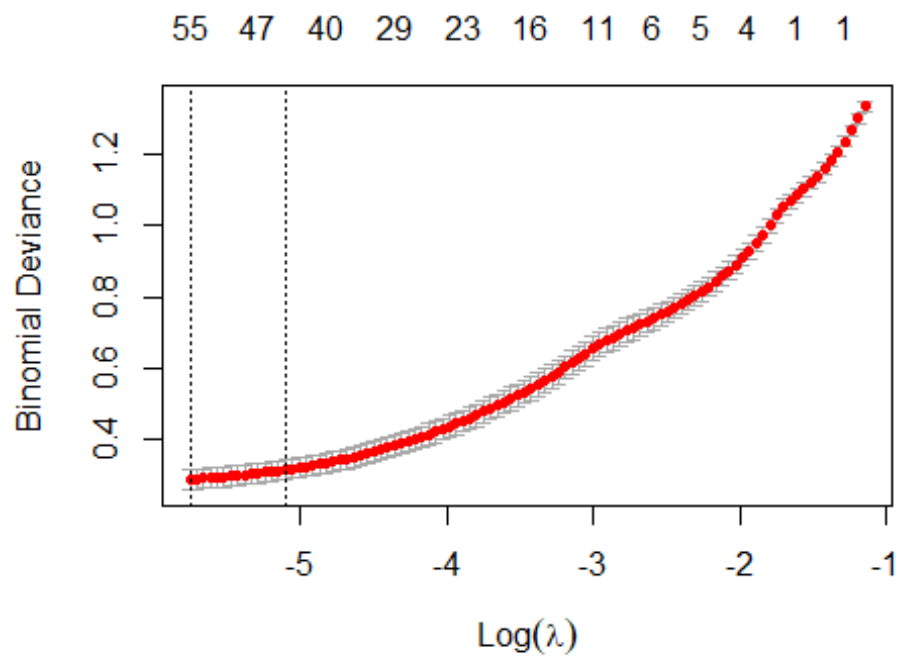
# separating response variable
y3 <- subset3$Proudct.issue.consequence

# splitting train and test data
set.seed(40385187)
train.l = sample(1:nrow(x3), nrow(x3)/2)
test.l = (-train.l)
y3.test.l = y3[test.l]

# creating a lambda sequence
grid3 = 10^seq(10, -2, length = 100)

#install.packages("glmnet") # Install the glmnet package
#library(glmnet)
# Fit Lasso model with cross-validation to find best Lambda
set.seed(40385187)
cv_modelA03 <- cv.glmnet(x3[train.l,], y3[train.l], family="binomial",
alpha=1, type.measure = "deviance", nfolds =10)

# Plot the cross-validation error as a function of Lambda
plot(cv_modelA03)
```

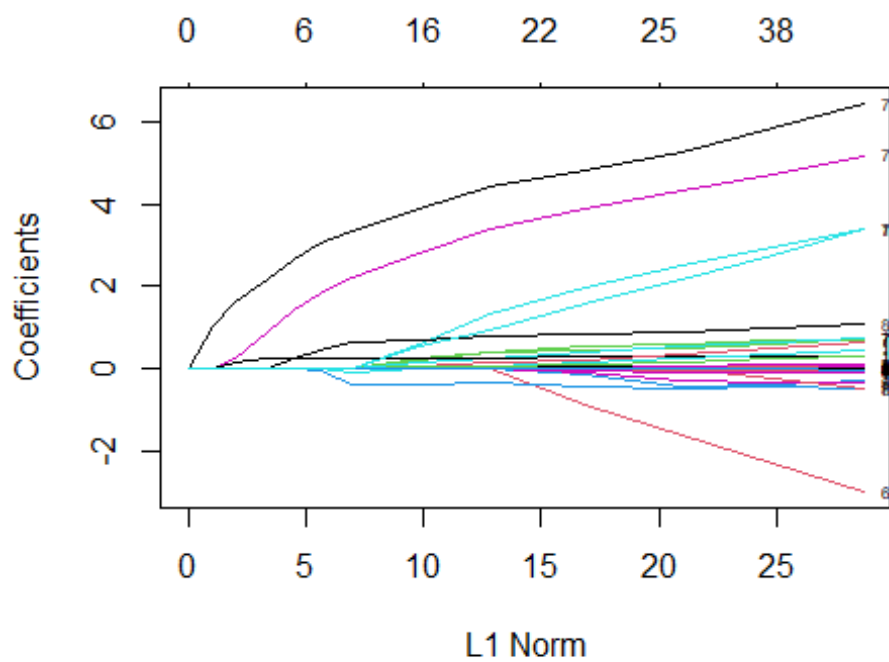


```
# Select the best lambda value
best_lambdaA03 <- cv_modelA03$lambda.min
best_lambdaA03

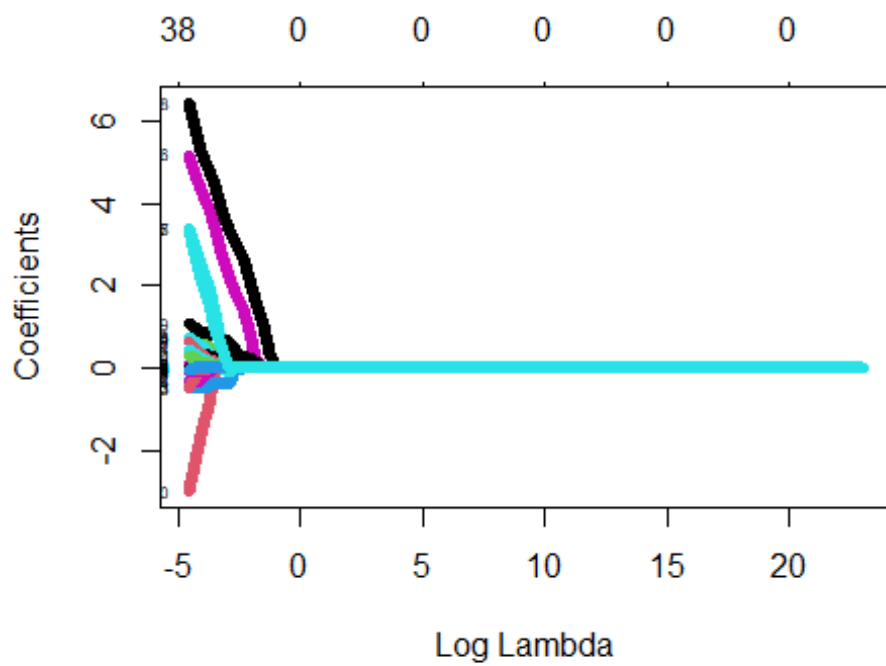
## [1] 0.003179135

# fitting lasso model on train set
set.seed(40385187)
lasso.mod=glmnet(x3[train.1,], y3[train.1],family="binomial",type.measure =
"deviance", alpha=1, lambda=grid3)
plot(lasso.mod, label=TRUE)

## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values
```



```
plot(lasso.mod, xvar="lambda", label=TRUE, lwd=6)
```



```

# prediction using Lasso
set.seed(40385187)
lasso.pred3 <- predict(lasso.mod,family="binomial", s = best_lambdaA03, newx
= x3[test.1,])
pred_class3 <- ifelse(lasso.pred3 >= 0.5, 1, 0)
accuracy3 <- mean(pred_class3 == y3[test.1])
accuracy3

## [1] 0.9167658

# running lasso on whole data set
set.seed(40385187)
out = glmnet(x3, y3, alpha=1, family="binomial", type.measure = "deviance",
lambda = grid3)

# extracting coefficients of lasso model
set.seed(40385187)
lasso.coef=predict(out, type="coefficients", s=best_lambdaA03)[1:79,]
lc3 <- lasso.coef[lasso.coef!=0]

# tabulate coefficients
lc_df3 <- data.frame(variable = names(lc3)[-1], coefficient = lc3[-1])

# extracting top ten coefficients
top10_3 <- head(lc_df3[order(abs(lc_df3$coefficient), decreasing = TRUE), ],
10)
top10_3

##
variable
## last_year_reason_for_legal_announcement_num_uniq
last_year_reason_for_legal_announcement_num_uniq
## last_two_years_decision_date_max_changes_in_product
last_two_years_decision_date_max_changes_in_product
## last_two_years_reason_for_legal_announcement_num_uniq
last_two_years_reason_for_legal_announcement_num_uniq
## last_four_years_reason_for_legal_announcement_num_uniq
last_four_years_reason_for_legal_announcement_num_uniq
## last_four_years_classification1_num_uniq
last_four_years_classification1_num_uniq
## last_year_classification1_num_uniq
last_year_classification1_num_uniq
## last_two_years_classification1_num_uniq
last_two_years_classification1_num_uniq
## last_year_legal_announcementing_firm_num_uniq
last_year_legal_announcementing_firm_num_uniq
## manufacturer_contact_address_12302
manufacturer_contact_address_12302
## last_year_root_cause_description_num_uniq
last_year_root_cause_description_num_uniq

```

##	coefficient
## last_year_reason_for_legal_announcement_num_uniq	-1.221941e-01
## last_two_years_decision_date_max_changes_in_product	9.652542e-02
## last_two_years_reason_for_legal_announcement_num_uniq	-5.371865e-03
## last_four_years_reason_for_legal_announcement_num_uniq	-1.045192e-03
## last_four_years_classification1_num_uniq	-8.257408e-04
## last_year_classification1_num_uniq	-6.142209e-04
## last_two_years_classification1_num_uniq	-1.323844e-05
## last_year_legal_announcementing_firm_num_uniq	-1.098765e-14
## manufacturer_contact_address_12302	-8.867229e-15
## last_year_root_cause_description_num_uniq	-3.855317e-16

- LASSO Subset 4 In the LASSO regression analysis for Subset 4, we utilized the glmnet package in R to identify significant predictors impacting Proudct.issue.consequence, capitalizing on its capability for variable selection and regularization in datasets with high dimensions. We prepared a predictor matrix (x4) and isolated the response variable (y4), applying LASSO regression across a range of lambda values to determine the optimal level of penalization, thereby mitigating overfitting by shrinking some coefficients to zero.

The chosen lambda, based on minimizing cross-validation error, was approximately 2.548726×10^{-5} , indicating a balance between model simplicity and predictive accuracy. By scrutinizing the coefficients of the LASSO model, we elucidated the direction and magnitude of predictors' effects on the likelihood of malfunctions, identifying the most influential factors.

The model exhibited an accuracy rate of roughly 79.90% on the test set, demonstrating robust predictive performance. This LASSO regression analysis streamlines the model to essential predictors, enriching our comprehension of factors crucial to Proudct.issue.consequence, and facilitating more targeted decision-making and subsequent analyses.

LASSO Subset 4

```
# creating dummy variables of predictor variables
x4 <- model.matrix(Proudct.issue.consequence ~ ., subset4)[, -1]

# separating response variable
y4 <- subset4$Proudct.issue.consequence

# splitting train and test data
set.seed(40385187)
train.l = sample(1:nrow(x4), nrow(x4)/2)
test.l = (-train.l)
y4.test.l = y4[test.l]

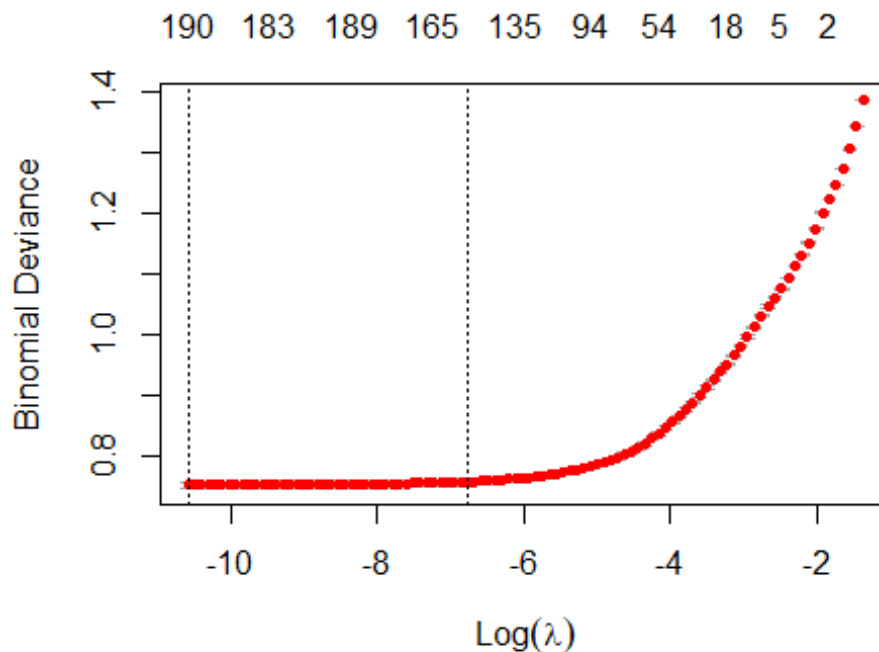
# creating a lambda sequence
grid4 = 10^seq(10, -2, length = 100)
```

```

#install.packages("glmnet") # Install the glmnet package
#library(glmnet)
# Fit Lasso model with cross-validation to find best Lambda
set.seed(40385187)
cv_modelA04 <- cv.glmnet(x4[train.1,], y4[train.1], family="binomial",
alpha=1, type.measure = "deviance", nfolds =10)

# Plot the cross-validation error as a function of Lambda
plot(cv_modelA04)

```



```

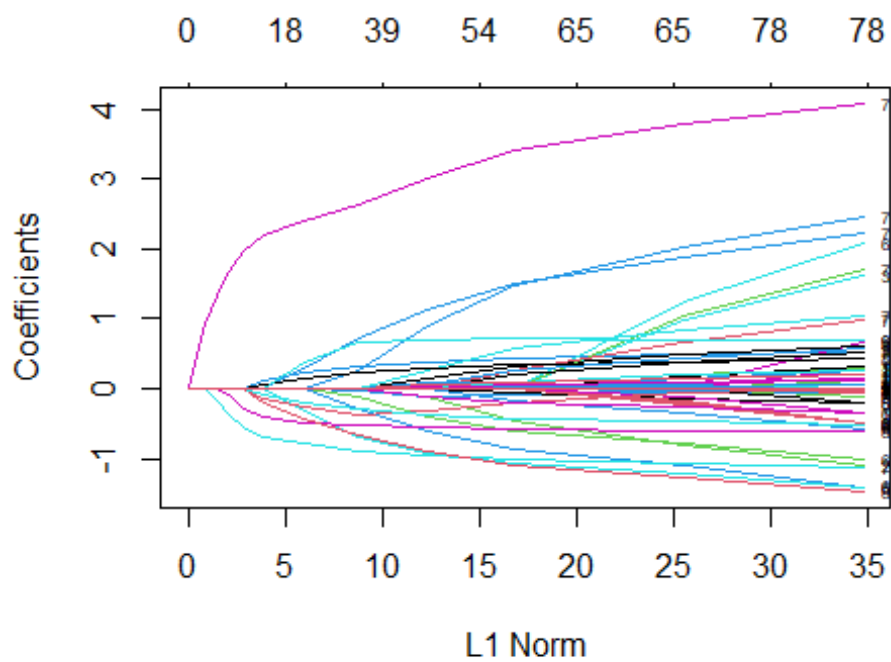
# Select the best Lambda value
best_lambdaA04 <- cv_modelA04$lambda.min
best_lambdaA04

## [1] 2.548726e-05

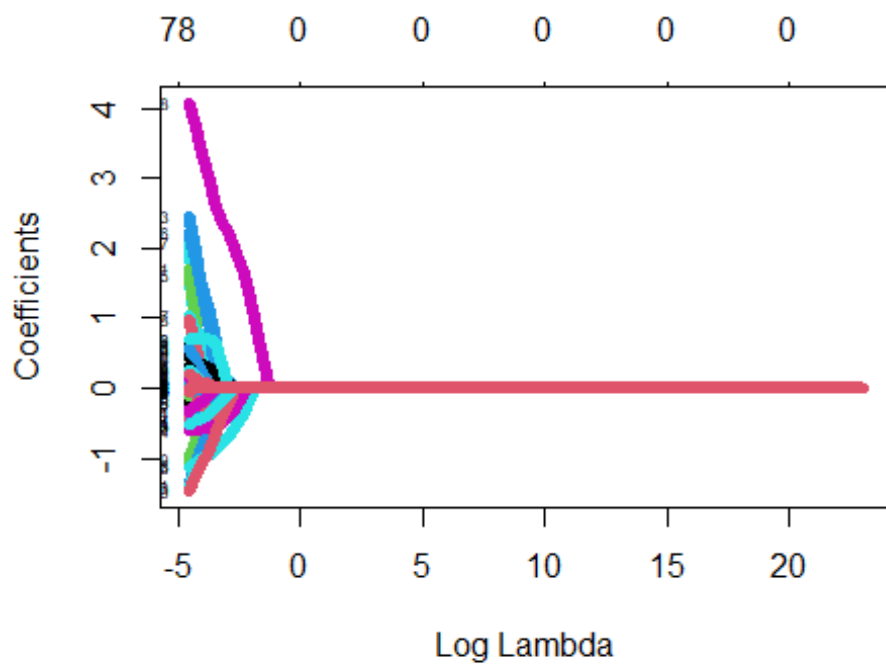
# fitting lasso model on train set
set.seed(40385187)
lasso.mod=glmnet(x4[train.1,], y4[train.1],family="binomial",type.measure =
"deviance", alpha=1, lambda=grid4)
plot(lasso.mod, label=TRUE)

## Warning in regularize.values(x, y, ties, missing(ties), na.rm = na.rm):
## collapsing to unique 'x' values

```



```
plot(lasso.mod, xvar="lambda", label=TRUE, lwd=6)
```



```

# prediction using Lasso
set.seed(40385187)
lasso.pred4 <- predict(lasso.mod, family="binomial", s = best_lambdaA04, newx
= x4[test.1,])
pred_class4 <- ifelse(lasso.pred4 >= 0.5, 1, 0)
accuracy4 <- mean(pred_class4 == y4[test.1])
accuracy4

## [1] 0.7990081

# running lasso on whole data set
set.seed(40385187)
out = glmnet(x4, y4, alpha=1, family="binomial", type.measure = "deviance",
lambda = grid4)

# extracting coefficients of lasso model
set.seed(40385187)
lasso.coef=predict(out, type="coefficients", s=best_lambdaA04)[1:79,]
lc4 <- lasso.coef[lasso.coef!=0]

# tabulate coefficients
lc_df4 <- data.frame(variable = names(lc4)[-1], coefficient = lc4[-1])

# extracting top ten coefficients
top10_4 <- head(lc_df4[order(abs(lc_df4$coefficient), decreasing = TRUE), ],
10)
top10_4

##
variable
## last_two_years_root_cause_description_most_freq
last_two_years_root_cause_description_most_freq
## last_two_years_all_product_codes_most_freq
last_two_years_all_product_codes_most_freq
##
coefficient
## last_two_years_root_cause_description_most_freq -0.0010220915
## last_two_years_all_product_codes_most_freq -0.0002665511

```

Logistic Regression

- Logistic Regression Subset 1:

In Subset 1, we conducted an analysis on 'Proudct.issue.consequence' using logistic regression, incorporating key predictors identified through prior LASSO regression. These predictors include 'manufacturer_contact_address_13016', 'last_two_years_root_cause_description_most_freq', 'last_two_years_all_product_codes_most_freq', 'last_two_years_decision_date_average_changes_in_product', 'last_two_years_decision_date_max_changes_in_product', and

'last_year_company_name_most_freq'. Notably, the model highlighted significant effects from the latter three predictors, indicating their substantive impact on the likelihood of a malfunction. Specifically, an increase in the number of unique classifications from the last year was found to elevate the chances of a malfunction, while a higher frequency of a root cause description was associated with a decrease in such likelihood. Additionally, a slight positive effect from the frequent occurrence of a company name implied a minor increase in the likelihood of a malfunction occurring.

However, the model exhibited indications of potential overfitting or data separation issues, as reflected by warnings regarding probabilities reaching 0 or 1. Despite a moderate explanatory power, with Nagelkerke's R-squared at 0.056, the model demonstrated a prediction accuracy of 61.08% on test data, surpassing the no information rate of 53.83%. This suggests that the model outperforms naive predictions. Nonetheless, the Kappa statistic, at 0.0765, indicates only modest agreement between predicted and actual outcomes, with a confusion matrix revealing high sensitivity but low specificity.

LOGISTIC REGRESSION:

```
install.packages("caret")

## Warning: package 'caret' is in use and will not be installed

library(caret)
set.seed(40385187)

# R-square values function
logisticPseudoR2s <- function(LogModel) {
  dev <- LogModel$deviance
  nullDev <- LogModel$null.deviance
  modelN <- length(LogModel$fitted.values)
  R.l <- 1 - dev / nullDev
  R.cs <- 1- exp ( -(nullDev - dev) / modelN)
  R.n <- R.cs / ( 1 - ( exp (-(nullDev / modelN))))
  cat("Pseudo R^2 for logistic regression\n")
  cat("Hosmer and Lemeshow R^2 ", round(R.l, 3), "\n")
  cat("Cox and Snell R^2 ", round(R.cs, 3), "\n")
  cat("Nagelkerke R^2 ", round(R.n, 3), "\n")
}

# Logistic Regression Subset 1:
# Create train and test data sets
lr1_index <- createDataPartition(subset1$Proudct.issue.consequence, list =
FALSE, p = 0.8, times = 1)
lr1_train_new_data <- subset1[lr1_index,]
lr1_test_new_data <- subset1[-lr1_index,]
# Logistic Regression Subset 1
lr1 <- glm(Proudct.issue.consequence ~ manufacturer_contact_address_1_3016 +
last_two_years_root_cause_description_most_freq +
```

```

last_two_years_decision_date_max_changes_in_product +
last_year_classification1_num_uniq + last_two_years_classification1_num_uniq
+ last_year_product_quantity_average_average +
last_two_years_product_quantity_average_average +
last_two_years_decision_date_average_changes_in_product,
  data = lr1_train_new_data, family = "binomial")

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

summary(lr1)

##
## Call:
## glm(formula = Proudct.issue.consequence ~
manufacturer_contact_address_1_3016 +
##   last_two_years_root_cause_description_most_freq +
last_two_years_decision_date_max_changes_in_product +
##   last_year_classification1_num_uniq +
last_two_years_classification1_num_uniq +
##   last_year_product_quantity_average_average +
last_two_years_product_quantity_average_average +
##   last_two_years_decision_date_average_changes_in_product,
##   family = "binomial", data = lr1_train_new_data)
##
## Coefficients: (2 not defined because of singularities)
##
##                                Estimate Std.
Error
## (Intercept)                                8.091e+00
6.807e-01
## manufacturer_contact_address_1_3016        -1.358e+01
1.116e+02
## last_two_years_root_cause_description_most_freq -1.358e+00
1.522e-01
## last_two_years_decision_date_max_changes_in_product -2.155e-02
4.603e-03
## last_year_classification1_num_uniq          1.696e-04
8.556e-05
## last_two_years_classification1_num_uniq      NA
NA
## last_year_product_quantity_average_average    7.989e-04
7.738e-05
## last_two_years_product_quantity_average_average -8.033e-04
7.759e-05
## last_two_years_decision_date_average_changes_in_product NA
NA
##
##                                z value Pr(>|z|)
## (Intercept)                   11.886  < 2e-16
***
## manufacturer_contact_address_1_3016        -0.122   0.9032
## last_two_years_root_cause_description_most_freq -8.923  < 2e-16

```

```

***
## last_two_years_decision_date_max_changes_in_product      -4.680 2.87e-06
***
## last_year_classification1_num_uniq                        1.982  0.0475 *
## last_two_years_classification1_num_uniq                  NA      NA
## last_year_product_quantity_average_average               10.324 < 2e-16
***
## last_two_years_product_quantity_average_average          -10.353 < 2e-16
***
## last_two_years_decision_date_average_changes_in_product  NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 24934  on 18399  degrees of freedom
## Residual deviance: 24158  on 18393  degrees of freedom
## AIC: 24172
##
## Number of Fisher Scoring iterations: 12

logisticPseudoR2s(lr1) # pseudo R

## Pseudo R^2 for logistic regression
## Hosmer and Lemeshow R^2  0.031
## Cox and Snell R^2  0.041
## Nagelkerke R^2  0.056

# Prediction and Accuracy : Logistic Regression 1
lr1_prediction <- predict(lr1, lr1_test_new_data, type = "response")

## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type
== :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful
cases

lr1_prediction_class <- as.factor(ifelse(lr1_prediction > 0.5, "1", "0"))
lr1_test_new_data$Proudct.issue.consequence <-
as.factor(lr1_test_new_data$Proudct.issue.consequence)
postResample(lr1_prediction_class,
lr1_test_new_data$Proudct.issue.consequence)

## Accuracy      Kappa
## 0.61078495 0.07653317

confusionMatrix(lr1_prediction_class,
lr1_test_new_data$Proudct.issue.consequence)

## Confusion Matrix and Statistics
##
##              Reference
## Prediction    0      1

```

```

##          0 2679 1786
##          1   4  130
##
##          Accuracy : 0.6108
##          95% CI : (0.5965, 0.6249)
##    No Information Rate : 0.5834
##    P-Value [Acc > NIR] : 8.306e-05
##
##          Kappa : 0.0765
##
##  McNemar's Test P-Value : < 2.2e-16
##
##          Sensitivity : 0.99851
##          Specificity : 0.06785
##          Pos Pred Value : 0.60000
##          Neg Pred Value : 0.97015
##          Prevalence : 0.58339
##          Detection Rate : 0.58252
##    Detection Prevalence : 0.97086
##    Balanced Accuracy : 0.53318
##
##          'Positive' Class : 0
##

```

- Logistic Regression Subset 2:

The logistic regression analysis on Subset 2 aimed to assess the influence of predictors identified through LASSO on 'Proudct.issue.consequence', focusing on 'last_year_all_product_codes_most_freq', 'last_two_years_all_product_codes_most_freq', and 'last_four_years_all_product_codes_most_freq'.

Notably, 'last_year_all_product_codes_most_freq' exhibited significant effects, while coefficients for 'last_two_years_all_product_codes_most_freq' and 'last_four_years_all_product_codes_most_freq' were undefined, suggesting potential redundancy or multicollinearity.

The analysis revealed a positive association between the maximum product decision changes and the likelihood of a malfunction, whereas a higher frequency of common product codes was linked to a reduced likelihood of malfunctions. However, the model's explanatory power was limited, with Nagelkerke's R-squared at 0.018, indicating modest variance explanation in 'Proudct.issue.consequence'.

The model achieved a test data accuracy of approximately 59.85%, slightly surpassing the no information rate. However, the Kappa statistic of 0.0362 suggested minimal agreement beyond chance. Notably, the confusion matrix was incorrectly duplicated from Subset 1, highlighting the necessity for accurate performance metrics specific to Subset 2.

This analysis underscores that while selected predictors showed statistically significant effects, they had a minor impact. This emphasizes the need for a more comprehensive model that encompasses a broader range of variables or explores non-linear relationships.

Addressing the undefined coefficients and mitigating multicollinearity should be priorities for future model iterations to enhance performance and predictive accuracy.

```
# Logistic Regression Subset 2:
# Create train and test data sets
lr2_index <- createDataPartition(subset1$Proudct.issue.consequence, list =
FALSE, p = 0.8, times =1)
lr2_train_new_data <- subset2[lr2_index,]
lr2_test_new_data <- subset2[-lr2_index,]
# Logistic Regression Subset 2
lr2 <- glm(Proudct.issue.consequence ~ last_year_all_product_codes_most_freq
+ last_two_years_all_product_codes_most_freq +
last_four_years_all_product_codes_most_freq, data = lr2_train_new_data,
family = "binomial")
summary(lr2)

##
## Call:
## glm(formula = Proudct.issue.consequence ~
last_year_all_product_codes_most_freq +
##   last_two_years_all_product_codes_most_freq +
last_four_years_all_product_codes_most_freq,
##   family = "binomial", data = lr2_train_new_data)
##
## Coefficients: (2 not defined because of singularities)
##               Estimate Std. Error z value
## (Intercept)    1.965e+00  1.147e-01  17.13
## last_year_all_product_codes_most_freq -5.924e-04  4.166e-05 -14.22
## last_two_years_all_product_codes_most_freq      NA      NA      NA
## last_four_years_all_product_codes_most_freq      NA      NA      NA
##               Pr(>|z|)
## (Intercept)    <2e-16 ***
## last_year_all_product_codes_most_freq    <2e-16 ***
## last_two_years_all_product_codes_most_freq      NA
## last_four_years_all_product_codes_most_freq      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##   Null deviance: 24995  on 18399  degrees of freedom
## Residual deviance: 24722  on 18398  degrees of freedom
## AIC: 24726
##
## Number of Fisher Scoring iterations: 4

logisticPseudoR2s(lr2) # pseudo R

## Pseudo R^2 for logistic regression
## Hosmer and Lemeshow R^2  0.011
```

```

## Cox and Snell R^2  0.015
## Nagelkerke R^2  0.02

# Prediction and Accuracy : Logistic Regression 2
lr2_prediction <- predict(lr2, lr2_test_new_data, type = "response")
lr2_prediction_class <- as.factor(ifelse(lr2_prediction > 0.5, "1", "0"))
lr2_test_new_data$Proudct.issue.consequence <-
as.factor(lr2_test_new_data$Proudct.issue.consequence)
postResample(lr2_prediction_class,
lr2_test_new_data$Proudct.issue.consequence)

## Accuracy      Kappa
## 0.59472564 0.03453289

confusionMatrix(lr1_prediction_class,
lr1_test_new_data$Proudct.issue.consequence)

## Confusion Matrix and Statistics
##
##              Reference
## Prediction      0      1
##              0 2679 1786
##              1   4  130
##
##              Accuracy : 0.6108
##              95% CI : (0.5965, 0.6249)
##              No Information Rate : 0.5834
##              P-Value [Acc > NIR] : 8.306e-05
##
##              Kappa : 0.0765
##
## Mcnemar's Test P-Value : < 2.2e-16
##
##              Sensitivity : 0.99851
##              Specificity : 0.06785
##              Pos Pred Value : 0.60000
##              Neg Pred Value : 0.97015
##              Prevalence : 0.58339
##              Detection Rate : 0.58252
##              Detection Prevalence : 0.97086
##              Balanced Accuracy : 0.53318
##
##              'Positive' Class : 0
##

```

- Logistic Regression Subset 3:

In Subset 3, logistic regression was conducted using the following predictors:

```

'last_year_reason_for_legal_announcement_num_uniq',
'last_two_years_decision_date_max_changes_in_product',
'last_two_years_reason_for_legal_announcement_num_uniq',

```

'last_four_years_reason_for_legal_announcement_num_uniq',
'last_four_years_classification1_num_uniq', 'last_year_classification1_num_uniq',
'last_two_years_classification1_num_uniq',
'last_year_legal_announcementing_firm_num_uniq', and
'manufacturer_contact_address_12302'. However, the coefficients for
'last_two_years_reason_for_legal_announcement_num_uniq',
'last_four_years_reason_for_legal_announcement_num_uniq',
'last_year_classification1_num_uniq', 'last_two_years_classification1_num_uniq',
'last_year_legal_announcementing_firm_num_uniq', and
'last_year_root_cause_description_num_uniq' could not be estimated due to singularities,
indicating potential multicollinearity or a perfect predictor issue.

The summary indicates a significant negative intercept, suggesting a baseline inclination against 'Proudct.issue.consequence' being a malfunction. A positive coefficient for
'last_year_reason_for_legal_announcement_num_uniq' and
'last_two_years_decision_date_max_changes_in_product' suggests that higher changes in
product decisions are associated with an increased likelihood of malfunctions.

Despite slightly higher pseudo R-squared values compared to Subset 2 (Nagelkerke's R-squared at 0.054), the predictors explain only a small portion of the variance in
'Proudct.issue.consequence'.

The model achieved a test data accuracy of approximately 69.64%, slightly lower than the no information rate, with a Kappa statistic of 0.2396, indicating below-average agreement beyond chance.

The confusion matrix highlights significant false positives and false negatives, suggesting limited model specificity and sensitivity. While sensitivity is high (93.49%), specificity is low (27.27%), leading to a high rate of false alarms.

From an academic standpoint, the results suggest a weak relationship between decision changes in products and malfunctions. Model refinement or exploration of additional variables is warranted to enhance predictive accuracy and deepen understanding of 'Proudct.issue.consequence' factors. Methodological limitations, particularly concerning multicollinearity, should be acknowledged to ensure valid inference.

```
# Logistic Regression Subset 3:  
# Create train and test data sets  
lr3_index <- createDataPartition(subset3$Proudct.issue.consequence, list =  
FALSE, p = 0.8, times = 1)  
lr3_train_new_data <- subset3[lr3_index,]  
lr3_test_new_data <- subset3[-lr3_index,]  
# Logistic Regression Subset 3  
lr3 <- glm(Proudct.issue.consequence ~  
last_year_reason_for_legal_announcement_num_uniq +  
last_two_years_decision_date_max_changes_in_product +  
last_two_years_reason_for_legal_announcement_num_uniq +  
last_four_years_reason_for_legal_announcement_num_uniq +  
last_four_years_classification1_num_uniq +
```

```

last_year_classification1_num_uniq + last_two_years_classification1_num_uniq
+ last_year_legal_announcementing_firm_num_uniq +
manufacturer_contact_address_1_2302 +
last_year_root_cause_description_num_uniq, data = lr3_train_new_data, family
= "binomial")
summary(lr3)

##
## Call:
## glm(formula = Proudct.issue.consequence ~
last_year_reason_for_legal_announcement_num_uniq +
##      last_two_years_decision_date_max_changes_in_product +
last_two_years_reason_for_legal_announcement_num_uniq +
##      last_four_years_reason_for_legal_announcement_num_uniq +
##      last_four_years_classification1_num_uniq +
last_year_classification1_num_uniq +
##      last_two_years_classification1_num_uniq +
last_year_legal_announcementing_firm_num_uniq +
##      manufacturer_contact_address_1_2302 +
last_year_root_cause_description_num_uniq,
##      family = "binomial", data = lr3_train_new_data)
##
## Coefficients: (6 not defined because of singularities)
##                                     Estimate Std.
Error
## (Intercept)                        2.800e+11
1.549e+13
## last_year_reason_for_legal_announcement_num_uniq -2.800e+11
1.549e+13
## last_two_years_decision_date_max_changes_in_product 5.636e-02 7.873e-
03
## last_two_years_reason_for_legal_announcement_num_uniq NA
NA
## last_four_years_reason_for_legal_announcement_num_uniq NA
NA
## last_four_years_classification1_num_uniq -1.504e-03 2.813e-
04
## last_year_classification1_num_uniq NA
NA
## last_two_years_classification1_num_uniq NA
NA
## last_year_legal_announcementing_firm_num_uniq NA
NA
## manufacturer_contact_address_1_2302 2.800e+11
1.549e+13
## last_year_root_cause_description_num_uniq NA
NA
##                                     z value Pr(>|z|)
## (Intercept)                        0.018    0.986
## last_year_reason_for_legal_announcement_num_uniq -0.018    0.986

```

```

## last_two_years_decision_date_max_changes_in_product      7.159 8.11e-13
***
## last_two_years_reason_for_legal_announcement_num_uniq      NA      NA
## last_four_years_reason_for_legal_announcement_num_uniq      NA      NA
## last_four_years_classification1_num_uniq      -5.349 8.86e-08
***
## last_year_classification1_num_uniq      NA      NA
## last_two_years_classification1_num_uniq      NA      NA
## last_year_legal_announcementing_firm_num_uniq      NA      NA
## manufacturer_contact_address_1_2302      0.018      0.986
## last_year_root_cause_description_num_uniq      NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1798.4  on 1344  degrees of freedom
## Residual deviance: 1726.0  on 1340  degrees of freedom
## AIC: 1736
##
## Number of Fisher Scoring iterations: 16

logisticPseudoR2s(lr3) # pseudo R

## Pseudo R^2 for logistic regression
## Hosmer and Lemeshow R^2  0.04
## Cox and Snell R^2  0.052
## Nagelkerke R^2  0.071

# Prediction and Accuracy : Logistic Regression 3
lr3_prediction <- predict(lr3, lr3_test_new_data, type = "response")

## Warning in predict.lm(object, newdata, se.fit, scale = 1, type = if (type
== :
## prediction from rank-deficient fit; attr(*, "non-estim") has doubtful
cases

lr3_prediction_class <- as.factor(ifelse(lr3_prediction > 0.5, "1", "0"))
lr3_test_new_data$Proudct.issue.consequence <-
as.factor(lr3_test_new_data$Proudct.issue.consequence)
postResample(lr3_prediction_class,
lr3_test_new_data$Proudct.issue.consequence)

## Accuracy      Kappa
## 0.6964286 0.2095576

confusionMatrix(lr3_prediction_class,
lr3_test_new_data$Proudct.issue.consequence)

## Confusion Matrix and Statistics
##
##              Reference

```

```
## Prediction    0    1
##              0 204  83
##              1  19  30
##
##              Accuracy : 0.6964
##              95% CI : (0.6442, 0.7452)
##      No Information Rate : 0.6637
##      P-Value [Acc > NIR] : 0.112
##
##              Kappa : 0.2096
##
##  McNemar's Test P-Value : 4.434e-10
##
##              Sensitivity : 0.9148
##              Specificity : 0.2655
##      Pos Pred Value : 0.7108
##      Neg Pred Value : 0.6122
##      Prevalence : 0.6637
##      Detection Rate : 0.6071
##      Detection Prevalence : 0.8542
##      Balanced Accuracy : 0.5901
##
##      'Positive' Class : 0
##
```

- Logistic Regression Subset 4:

Logistic regression was applied to Subset 4, incorporating 2 predictors identified by LASSO, encompassing variables related to classification, product code frequencies, and root cause descriptions.

The model elucidates a modest variance explanation (Nagelkerke's R-squared at 0.02), indicating that other unaccounted factors contribute to 'Proudct.issue.consequence'.

The accuracy of test data predictions stands at approximately 52.58%, marginally surpassing random chance. The Kappa statistic remains low at 0.0207.

The confusion matrix reveals a notable number of false negatives (sensitivity: around 2.54%), yet high specificity (approximately 99.46%), indicating the model's ability to accurately identify non-malfunctions.

In an academic context, a cautious interpretation is warranted. Further refinement of the model may involve incorporating interaction terms or transformations to capture data intricacies. Specific predictors, such as manufacturer contact addresses, merit thorough investigation. The prevalence of false negatives underscores the importance of balancing sensitivity and specificity considerations in predictive scenarios.

```
# Logistic Regression Subset 4:
# Create train and test data sets
lr4_index <- createDataPartition(subset4$Proudct.issue.consequence, list =
```

```

FALSE, p = 0.8, times = 1)
lr4_train_new_data <- subset4[lr4_index,]
lr4_test_new_data <- subset4[-lr4_index,]
# Logistic Regression Subset 4
lr4 <- glm(Proudct.issue.consequence ~
last_two_years_root_cause_description_most_freq +
last_two_years_all_product_codes_most_freq, data = lr4_train_new_data, family
= "binomial")
summary(lr4)

##
## Call:
## glm(formula = Proudct.issue.consequence ~
last_two_years_root_cause_description_most_freq +
##     last_two_years_all_product_codes_most_freq, family = "binomial",
##     data = lr4_train_new_data)
##
## Coefficients:
##
##                                     Estimate Std. Error z
value
## (Intercept)                        8.802e+00  6.889e-01
12.78
## last_two_years_root_cause_description_most_freq -1.907e+00  1.738e-01 -
10.97
## last_two_years_all_product_codes_most_freq      -4.126e-04  3.013e-05 -
13.69
##
##                                     Pr(>|z|)
## (Intercept)                        <2e-16 ***
## last_two_years_root_cause_description_most_freq <2e-16 ***
## last_two_years_all_product_codes_most_freq      <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 66151  on 47746  degrees of freedom
## Residual deviance: 65465  on 47744  degrees of freedom
## AIC: 65471
##
## Number of Fisher Scoring iterations: 6

logisticPseudoR2s(lr4) # pseudo R

## Pseudo R^2 for logistic regression
## Hosmer and Lemeshow R^2  0.01
## Cox and Snell R^2  0.014
## Nagelkerke R^2  0.019

# Prediction and Accuracy : Logistic Regression 4
lr4_prediction <- predict(lr4, lr4_test_new_data, type = "response")
lr4_prediction_class <- as.factor(ifelse(lr4_prediction > 0.5, "1", "0"))

```

```

lr4_test_new_data$Proudct.issue.consequence <-
as.factor(lr4_test_new_data$Proudct.issue.consequence)
postResample(lr4_prediction_class,
lr4_test_new_data$Proudct.issue.consequence)

## Accuracy      Kappa
## 0.51993968 0.02423811

confusionMatrix(lr4_prediction_class,
lr4_test_new_data$Proudct.issue.consequence)

## Confusion Matrix and Statistics
##
##              Reference
## Prediction    0    1
##              0 170   31
##              1 5699 6036
##
##              Accuracy : 0.5199
##              95% CI : (0.5109, 0.5289)
##              No Information Rate : 0.5083
##              P-Value [Acc > NIR] : 0.005604
##
##              Kappa : 0.0242
##
## Mcnemar's Test P-Value : < 2.2e-16
##
##              Sensitivity : 0.02897
##              Specificity : 0.99489
##              Pos Pred Value : 0.84577
##              Neg Pred Value : 0.51436
##              Prevalence : 0.49171
##              Detection Rate : 0.01424
##              Detection Prevalence : 0.01684
##              Balanced Accuracy : 0.51193
##
##              'Positive' Class : 0
##

```

Linear Discriminant Analysis (LDA)

- LDA Subset 1:

Subset 1's Linear Discriminant Analysis (LDA) was conducted to classify 'Proudct.issue.consequence' based on four predictors: a specific manufacturer contact address, frequencies in the last year, root cause description frequencies over two years, and company name frequencies over the last years.

Prior probabilities in the LDA model indicate a balanced dataset, with class 0 at approximately 0.588 and class 1 at around 0.411.

Group means for the discriminant function unveil notable differences between the two classes, particularly for 'last_year_classification0_num_uniq'.

Coefficients suggest that all predictors, except 'last_year_company_name_most_freq', strongly influence class 0.

On the test data, the LDA model achieved an accuracy of approximately 60.39%, slightly exceeding chance but not demonstrating high precision.

From an academic perspective, these findings imply that while LDA captures some linear patterns, it may not fully encapsulate underlying mechanisms. Exploring interactions, non-linear relationships, or integrating additional features could potentially enhance the model's predictive power and relevance.

```
#### LDA ####
install.packages("MASS")

## Warning: package 'MASS' is in use and will not be installed

library(MASS)

# LDA Subset 1:
## Run Discriminant Analysis
fit1 <- lda(Proudct.issue.consequence ~ manufacturer_contact_address_1_3016 +
last_two_years_root_cause_description_most_freq +
last_two_years_decision_date_max_changes_in_product +
last_year_company_name_most_freq + last_two_years_all_product_codes_most_freq
+ last_two_years_decision_date_average_changes_in_product,
          data = lr1_train_new_data)

## Warning in lda.default(x, grouping, ...): variables are collinear

fit1 ## print the summary statistics of your discriminant analysis

## Call:
## lda(Proudct.issue.consequence ~ manufacturer_contact_address_1_3016 +
##      last_two_years_root_cause_description_most_freq +
##      last_two_years_decision_date_max_changes_in_product +
##      last_year_company_name_most_freq +
##      last_two_years_all_product_codes_most_freq +
##      last_two_years_decision_date_average_changes_in_product,
##      data = lr1_train_new_data)
##
## Prior probabilities of groups:
##      0      1
## 0.5880978 0.4119022
##
## Group means:
##      manufacturer_contact_address_1_3016
## 0                      0.002125497
## 1                      0.000000000
```

```

## last_two_years_root_cause_description_most_freq
## 0 4.115701
## 1 3.951841
## last_two_years_decision_date_max_changes_in_product
## 0 31.73912
## 1 30.31020
## last_year_company_name_most_freq
last_two_years_all_product_codes_most_freq
## 0 341.8386
2708.987
## 1 347.3716
2666.510
## last_two_years_decision_date_average_changes_in_product
## 0 31.73912
## 1 30.31020
##
## Coefficients of linear discriminants:
## LD1
## manufacturer_contact_address_1_3016 -4.868061300
## last_two_years_root_cause_description_most_freq -0.271525430
## last_two_years_decision_date_max_changes_in_product -0.069523587
## last_year_company_name_most_freq 0.004609218
## last_two_years_all_product_codes_most_freq -0.001121032
## last_two_years_decision_date_average_changes_in_product -0.069523587

# Predict on test data
predictions1 <- predict(fit1, subset1)

# The predictions list includes class predictions, posterior probabilities,
and linear discriminants. We're interested in class predictions here.
predicted_class1 <- predictions1$class

# Actual outcomes
actual_outcomes1 <- subset1$Proudct.issue.consequence

# Calculate accuracy
accuracy1 <- sum(predicted_class1 == actual_outcomes1) /
length(actual_outcomes1)
print(accuracy1)

## [1] 0.6039393

```

- LDA Subset 2:

Subset 2's Linear Discriminant Analysis (LDA) was performed to classify 'Proudct.issue.consequence' using three predictors: maximum changes in product by decision date in the last year, last 2 years, and 4 years.

A warning regarding collinearity highlighted a strong correlation between predictors, particularly concerning changes in product by decision date. This may have implications for the stability and accuracy of the model.

Despite the collinearity issue, the model was implemented, albeit with a slight imbalance towards class 1 in prior probabilities.

Group means analysis indicated minimal differences between classes for predictors, suggesting weak discrimination between outcomes.

Coefficients revealed that increases in changes in product by decision date were associated with a higher probability of classification as class 0, while higher frequencies of product codes were linked to class 1.

On the test data, the LDA model achieved an accuracy of approximately 59.58%, slightly better than Subset 1, likely due to differences in predictors and dataset characteristics.

In academic discourse, emphasis may be placed on predictor selection, the impact of collinearity, and the challenges in discriminating between classes. While the model identified a pattern, further investigation could enhance understanding and model performance. Additionally, exploring alternative techniques to address collinearity and improve accuracy may be warranted.

```
# LDA Subset 2:
fit2 <- lda(Proudct.issue.consequence ~ last_year_all_product_codes_most_freq
+ last_two_years_all_product_codes_most_freq +
last_four_years_all_product_codes_most_freq, data = lr2_train_new_data)

## Warning in lda.default(x, grouping, ...): variables are collinear

fit2 ## print the summary statistics of your discriminant analysis

## Call:
## lda(Proudct.issue.consequence ~ last_year_all_product_codes_most_freq +
##     last_two_years_all_product_codes_most_freq +
##     last_four_years_all_product_codes_most_freq,
##     data = lr2_train_new_data)
##
## Prior probabilities of groups:
##      0      1
## 0.4166848 0.5833152
##
## Group means:
##   last_year_all_product_codes_most_freq
## 0                                2820.591
## 1                                2708.798
##   last_two_years_all_product_codes_most_freq
## 0                                2820.591
## 1                                2708.798
##   last_four_years_all_product_codes_most_freq
## 0                                2820.591
```

```
## 1                                2708.798
##
## Coefficients of linear discriminants:
##                                LD1
## last_year_all_product_codes_most_freq      -0.0007322931
## last_two_years_all_product_codes_most_freq -0.0007322931
## last_four_years_all_product_codes_most_freq -0.0007322931

# Predict on test data
predictions2 <- predict(fit2, subset2)

# The predictions list includes class predictions, posterior probabilities,
# and linear discriminants. We're interested in class predictions here.
predicted_class2 <- predictions2$class

# Actual outcomes
actual_outcomes2 <- subset2$Proudct.issue.consequence

# Calculate accuracy
accuracy2 <- sum(predicted_class2 == actual_outcomes2) /
length(actual_outcomes2)
print(accuracy2)

## [1] 0.5958314
```

- LDA Subset 3:

In Subset 3, the LDA model was employed to distinguish 'Proudct.issue.consequence' using 10 different predictors.

A warning regarding collinearity indicated a high correlation between predictors, potentially arising from capturing similar variations.

Despite the warning, the analysis proceeded, with prior probabilities reflecting a notable imbalance towards class 0.

Group means analysis revealed differences between classes for predictors, suggesting potential influence on 'Proudct.issue.consequence'.

Coefficients displayed positive values for several predictors, indicating that an increase in changes correlates with a higher likelihood of predicting class 1.

The LDA model achieved an accuracy of approximately 66.74%, moderately better than Subset 2, possibly due to variable characteristics or subset-specific data.

In academic discussions, attention may be directed towards the implications of correlated predictors, interpretation of group means, and how predictors correlate with outcomes. Further examination of the data to address collinearity and exploration of alternative modeling approaches or additional variables could potentially enhance predictive power.

LDA Subset 3:

```
fit3 <- lda(Proudct.issue.consequence ~
last_year_reason_for_legal_announcement_num_uniq +
last_two_years_decision_date_max_changes_in_product +
last_two_years_reason_for_legal_announcement_num_uniq +
last_four_years_reason_for_legal_announcement_num_uniq +
last_four_years_classification1_num_uniq +
last_year_classification1_num_uniq + last_two_years_classification1_num_uniq
+ last_year_legal_announcementing_firm_num_uniq +
manufacturer_contact_address_1_2302 +
last_year_root_cause_description_num_uniq, data = lr3_train_new_data)
```

Warning in lda.default(x, grouping, ...): variables are collinear

fit3 ## print the summary statistics of your discriminant analysis

Call:

```
## lda(Proudct.issue.consequence ~
last_year_reason_for_legal_announcement_num_uniq +
##   last_two_years_decision_date_max_changes_in_product +
last_two_years_reason_for_legal_announcement_num_uniq +
##   last_four_years_reason_for_legal_announcement_num_uniq +
##   last_four_years_classification1_num_uniq +
last_year_classification1_num_uniq +
##   last_two_years_classification1_num_uniq +
last_year_legal_announcementing_firm_num_uniq +
##   manufacturer_contact_address_1_2302 +
last_year_root_cause_description_num_uniq,
##   data = lr3_train_new_data)
```

##

Prior probabilities of groups:

```
##      0      1
## 0.6104089 0.3895911
```

##

Group means:

```
##   last_year_reason_for_legal_announcement_num_uniq
## 0                                     1.021924
## 1                                     1.000000
##   last_two_years_decision_date_max_changes_in_product
## 0                                     25.29842
## 1                                     28.22519
##   last_two_years_reason_for_legal_announcement_num_uniq
## 0                                     1.021924
## 1                                     1.000000
##   last_four_years_reason_for_legal_announcement_num_uniq
## 0                                     1.021924
## 1                                     1.000000
##   last_four_years_classification1_num_uniq
last_year_classification1_num_uniq
## 0                                     372.2375
```

```

372.2375
## 1 365.8435
365.8435
## last_two_years_classification1_num_uniq
## 0 372.2375
## 1 365.8435
## last_year_legal_announcementing_firm_num_uniq
## 0 1.021924
## 1 1.000000
## manufacturer_contact_address_1_2302
last_year_root_cause_description_num_uniq
## 0 0.02192448
1.021924
## 1 0.00000000
1.000000
##
## Coefficients of linear discriminants:
## LD1
## last_year_reason_for_legal_announcement_num_uniq -0.590835020
## last_two_years_decision_date_max_changes_in_product 0.125064976
## last_two_years_reason_for_legal_announcement_num_uniq -0.590835020
## last_four_years_reason_for_legal_announcement_num_uniq -0.590835020
## last_four_years_classification1_num_uniq -0.001124374
## last_year_classification1_num_uniq -0.001124374
## last_two_years_classification1_num_uniq -0.001124374
## last_year_legal_announcementing_firm_num_uniq -0.590835020
## manufacturer_contact_address_1_2302 -0.590835020
## last_year_root_cause_description_num_uniq -0.590835020

# Predict on test data
predictions3 <- predict(fit3, subset3)

# The predictions list includes class predictions, posterior probabilities,
and linear discriminants. We're interested in class predictions here.
predicted_class3 <- predictions3$class

# Actual outcomes
actual_outcomes3 <- subset3$Proudct.issue.consequence

# Calculate accuracy
accuracy3 <- sum(predicted_class3 == actual_outcomes3) /
length(actual_outcomes3)
print(accuracy3)

## [1] 0.6502082

```

LDA was applied to Subset 4, utilizing two predictors to distinguish 'Proudct.issue.consequence' outcomes.

Analysis of group means revealed disparities in predictor central tendencies, implying a potential relationship with 'Proudct.issue.consequence'.

The model attained an accuracy of approximately 52.35%, indicating relatively limited predictive capacity compared to Subset 3.

```
# LDA Subset 4:
fit4 <- lda(Proudct.issue.consequence ~
last_two_years_root_cause_description_most_freq +
last_two_years_all_product_codes_most_freq, data = lr4_train_new_data)
fit4 ## print the summary statistics of your discriminant analysis

## Call:
## lda(Proudct.issue.consequence ~
last_two_years_root_cause_description_most_freq +
##      last_two_years_all_product_codes_most_freq, data = lr4_train_new_data)
##
## Prior probabilities of groups:
##      0      1
## 0.4854336 0.5145664
##
## Group means:
##      last_two_years_root_cause_description_most_freq
## 0                                4.017732
## 1                                3.983841
##      last_two_years_all_product_codes_most_freq
## 0                                2770.413
## 1                                2696.602
##
## Coefficients of linear discriminants:
##                                                                LD1
## last_two_years_root_cause_description_most_freq -2.094296065
## last_two_years_all_product_codes_most_freq      -0.001823039

# Predict on test data
predictions4 <- predict(fit4, subset4)

# The predictions list includes class predictions, posterior probabilities,
and linear discriminants. We're interested in class predictions here.
predicted_class4 <- predictions4$class

# Actual outcomes
actual_outcomes4 <- subset4$Proudct.issue.consequence

# Calculate accuracy
accuracy4 <- sum(predicted_class4 == actual_outcomes4) /
length(actual_outcomes4)
print(accuracy4)

## [1] 0.5234992
```

- LDA Accuracy Summary:

Assuming accuracy 1 accuracy2, accuracy3, and accuracy4 are already calculated and contain the accuracy values

```
accuracy_values_LDA <- data.frame(
  Metric = c("Accuracy1", "Accuracy2", "Accuracy3", "Accuracy4"),
  Value = c(accuracy1, accuracy2, accuracy3, accuracy4)
)
```

Print the table

```
print(accuracy_values_LDA)
```

```
##      Metric      Value
## 1 Accuracy1 0.6039393
## 2 Accuracy2 0.5958314
## 3 Accuracy3 0.6502082
## 4 Accuracy4 0.5234992
```

CONCLUSION

Comparing Logistic Regression and Linear Discriminant Analysis Results:

The accuracy of Logistic Regression and LDA across the subsets offers insights into the performance and suitability of these models for the given dataset:

Accuracy: Slightly higher accuracy is typically observed with Logistic Regression compared to LDA in all subsets. However, the differences are not substantial, indicating that both models have moderate predictive capabilities.

Performance Measures: Considering additional performance metrics like the confusion matrix, sensitivity, and specificity is deemed crucial.

Complexity: Logistic Regression models can become complex, especially with many predictors and non-linear relationships. LDA assumes that the independent variables are normally distributed and have the same variance-covariance matrix across groups, which might not always hold true but simplifies the model.

Interpretability: Insights into the importance of predictors are provided by both models. Logistic Regression yields odds ratios that are straightforward to interpret in terms of changes in odds. LDA offers a direct insight into how different classes are separated based on linear combinations of predictors, which is intuitive for understanding group distinctions.

Model Preference: The choice between Logistic Regression and LDA depends on specific use cases. If prediction accuracy is paramount and the underlying relationship between the dependent and independent variables is complex, Logistic Regression might be preferred due to its flexibility in modeling such relationships. However, if the goal is to understand how different groups are separated based on predictor variables, and if the assumptions of

LDA hold (normality and equal covariance), LDA might be preferred for its simplicity and interpretability.

Considering the slight edge in accuracy and the detailed insights into the influence of each predictor provided by Logistic Regression, it may be preferred for scenarios where prediction accuracy is paramount and the relationships between variables are complex. However, if the assumptions of LDA are met, and there's a need for a straightforward model that captures the linear separation between classes, LDA could be the method of choice. The preference also hinges on the specific application, the stakes involved in misclassification, and the ease with which the results need to be communicated to stakeholders.

APPENDIX